

SYLLABUS

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State v. Darryl Nieves (A-26/27-23) (088683)

Argued October 21, 2024 -- Decided November 20, 2025

JUSTICE PIERRE-LOUIS, writing for the Court.

In this appeal, the Court considers whether expert testimony regarding Shaken Baby Syndrome/Abusive Head Trauma (SBS/AHT) is sufficiently reliable to go before a jury in two separate cases -- State v. Nieves and State v. Cifelli.

In both matters, the young children exhibited symptoms that have come to be associated with SBS/AHT and referred to as the “triad of symptoms” -- subdural hematoma, retinal hemorrhages, and encephalopathy. In both cases, the same doctor determined that the children were victims of child abuse, specifically SBS/AHT. Because the children were in the care of their fathers when they began to exhibit the above-mentioned symptoms, both men were charged with criminal offenses.

The State sought to present the expert testimony of a doctor that the only explanation for the children’s symptoms, to a reasonable degree of medical certainty, was that the children were shaken by the caregiver. The defense in both cases moved to exclude the testimony, challenging the scientific basis and reliability of the theory that shaking alone, without some other impact to the head, can cause the symptoms associated with SBS/AHT.

After a hearing to explore the admissibility of the evidence in Nieves, reviewed in detail on pages 29-57 of the Court’s opinion, the trial court excluded SBS/AHT testimony from the trial. The trial court in Cifelli followed suit. The appeals were consolidated and the Appellate Division affirmed. The Court granted certification in Nieves and leave to appeal in Cifelli. 256 N.J. 451 (2024).

HELD: The Court agrees with the trial courts and Appellate Division that the State has not met its burden in establishing the reliability of SBS/AHT testimony here.

1. The Court reviews the history of SBS/AHT over the past six decades (pp. 9-26):

- In 1968, neurosurgeon Dr. Ayub Ommaya conducted an experimental study focused specifically on whiplash injuries from car accidents. Although Dr. Ommaya’s study did not concern head trauma in infants from shaking, it

became a foundation for diagnoses of SBS/AHT, particularly its conclusion that brain injuries could occur by rotational displacement of the head on the neck alone, without significant direct head impact.

- In 1971, neurosurgeon Dr. Arthur Norman Guthkelch presented the theory of SBS, partially relying on Dr. Ommaya's 1968 study. Dr. Guthkelch hypothesized that an "infant having been shaken rather than struck by its parent" might sustain a subdural hematoma.
- In 1972 and 1974, pediatric radiologist Dr. John Caffey published papers on "whiplash-shaking" of infants, relying on historical cases and Dr. Ommaya's study. In comparing the force of infant shaking with the force generated during a car accident whiplash event, Dr. Caffey stated -- without elaboration or citation -- that manual shaking, when repeated, "may be much more harmful to the brain" and "the veins in the eyes" than a car accident whiplash event" and that "[c]urrent evidence, though manifestly incomplete and largely circumstantial, warrants a nationwide educational campaign" on the potential risks of whiplash shaking of infants. In the years following Dr. Caffey's 1974 study, the theory of whiplash infant shaking syndrome, later referred to as shaken baby syndrome, began to gain traction in the medical community.
- In 1987, Dr. Ann-Christine Duhaime, using infant models, conducted the first biomechanical study testing shaken baby syndrome and the effects of shaking without impact. Dr. Duhaime concluded that shaking alone does not produce shaken baby syndrome.
- In 2002, Dr. Ommaya, whose whiplash study was the basis for Dr. Guthkelch's conclusions and Dr. Caffey's whiplash shaken infant syndrome hypothesis, published a study criticizing other researchers' reliance on his own study as the scientific foundation of SBS. A decade later, Dr. Guthkelch also questioned the science behind SBS.
- In 2018, while the SBS/AHT debate continued, a group of physicians and pediatric radiologists published a Consensus Statement on AHT, which the American Academy of Pediatrics later endorsed. The Consensus Statement cited to the studies by Dr. Caffey and noted Dr. Caffey's citation to Dr. Ommaya and Dr. Guthkelch. It acknowledged Dr. Duhaime's conclusion that shaking alone cannot generate the force needed to cause the injuries that result in SBS/AHT cases but noted that contrary evidence -- confessions by caregivers -- supports the argument that shaking alone can cause SBS/AHT. The Consensus Statement also noted that SBS/AHT is a medical diagnosis, not a legal finding of murder, and declared that "[t]here is no controversy concerning the medical validity of the existence of AHT."

2. For expert testimony to be admissible under N.J.R.E. 702, its proponent must establish that (1) the subject matter of the testimony is beyond the ken of the average juror; (2) the field of inquiry is at a state of the art such that an expert's testimony could be sufficiently reliable; and (3) the witness has sufficient expertise to offer the testimony. During the relevant period, New Jersey courts relied on the standard set forth in Frye v. United States, 293 F. 1013 (D.C. Cir. 1923), to assess the reliability prong. The Frye standard requires trial courts to determine whether the science underlying the proposed expert testimony has gained general acceptance in the particular field in which it belongs. In certain circumstances, there might be more than one relevant scientific community to consider. In Olenowski II, for example, the Court recognized that the relevant scientific communities in determining the reliability of Drug Recognition Expert testimony included both medicine and toxicology. State v. Olenowski (Olenowski II), 255 N.J. 529, 604 (2023). Further, courts may revisit previously accepted theories as the science develops. In State v. J.L.G., for example, the Court determined that certain expert testimony regarding Child Sexual Abuse Accommodation Syndrome no longer met the reliability prong and was therefore inadmissible in criminal trials because it was based on clinical practice and not supported by objectively tested scientific evidence. 234 N.J. 265, 272, 291 (2018). (pp. 78-83)

3. Few New Jersey court opinions have addressed the reliability of SBS/AHT, and none involve the reliability of SBS/AHT in the context of shaking without impact. However, many cases from around the country in the past 15 years have focused on the reliability of SBS/AHT, and the Court reviews such cases. (pp. 83-91)

4. Turning to the matters before it, the Court first determines the relevant scientific community or communities for purposes of SBS/AHT. As the State's expert testified, the starting point in the evolution of SBS/AHT was Dr. Ommaya's 1968 whiplash study, on which Drs. Guthkelch and Caffey later relied in conceptualizing SBS/AHT. It is therefore evident that the foundation of SBS/AHT lies in biomechanical science and engineering. A scientific community is either relevant or not for purposes of determining admissibility of scientific evidence at trial -- degrees of relevance are not weighed. As in Olenowski II, there can certainly be more than one relevant scientific community for purposes of Frye. Here, the relevant scientific communities for purposes of determining the reliability of SBS/AHT expert testimony are both the medical/pediatric community and the biomechanical engineering community. (pp. 92-94)

5. The Court next considers whether, under the Frye standard of reliability, the State has met its burden of establishing that its expert's testimony regarding SBS/AHT without impact is sufficiently reliable for admission at trial. Based on the testimony, evidence, and scientific studies and writings, the State has not met its burden of clearly establishing general acceptance in the relevant scientific communities

because it has not shown general acceptance within the biomechanical community regarding whether shaking without impact can produce the “triad” of symptoms associated with SBS/AHT. The record established in the Frye hearing regarding the challenges to the science behind SBS/AHT in these matters questions the scientific and biomechanical underpinnings of the diagnosis of SBS/AHT as the sole basis for a legal finding of child abuse. As the trial court found, the evidence presented at the Frye hearing -- including the testimony by the State’s single expert witness -- showed that there was no test supporting a finding that humans can produce the physical force necessary to cause the symptoms associated with SBS/AHT in a child. There is evidence of general acceptance by many in the medical community, but the State must also establish general acceptance in the biomechanical community, and it has failed to do so. (pp. 94-107)

6. The Court explains that if there is physical evidence of trauma to a child or other evidence of abuse, the State can present such evidence to a jury. And if new, reliable, scientific evidence is developed, the State can, in a future case, make a showing under the Daubert standard adopted in State v. Olenowski (Olenowski I), 253 N.J. 133, 139, 155 (2023), that expert testimony regarding SBS/AHT without impact is reliable. In such a case, scientific evidence and research, both old and new, could be presented and considered. (p. 108)

AFFIRMED.

JUSTICE WAINER APTER, dissenting, stresses that SBS/AHT has been explicitly affirmed by every major discipline involved in its diagnosis and treatment and expresses the view that the Court here substitutes its judgment for that of the relevant scientific community. Justice Wainer Apter writes that, in the few prior instances in which the Court determined that evidence did not satisfy the Frye standard, the evidence lacked the endorsement of even one major scientific association or society, whereas SBS/AHT is endorsed by all of them. Further, Justice Wainer Apter notes, every other state admits expert testimony diagnosing SBS/AHT, and every other court that has considered the question has held that such evidence is admissible. Justice Wainer Apter writes that the majority has allowed individual biomechanical engineers to veto the consensus perspective of every major medical society in the world -- in a case about the admissibility of a medical diagnosis. Justice Wainer Apter would leave it to the jury to weigh the experts’ competing testimony.

CHIEF JUSTICE RABNER and JUSTICES PATTERSON, FASCIALE, NORIEGA, and HOFFMAN join in JUSTICE PIERRE-LOUIS’s opinion. JUSTICE WAINER APTER filed a dissent.

SUPREME COURT OF NEW JERSEY

A-26/27 September Term 2023

088683

State of New Jersey,

Plaintiff-Appellant,

v.

Darryl Nieves,

Defendant-Respondent.

State of New Jersey,

Plaintiff-Appellant,

v.

Michael Cifelli,

Defendant-Respondent.

On certification to and appeal from the Superior
Court, Appellate Division, whose opinion is reported
at 476 N.J. Super. 609 (App. Div. 2023).

Argued
October 21, 2024

Decided
November 20, 2025

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Lawrence S. Lustberg submitted a brief on behalf of amici curiae Lindsay “Dutch” Johnson, Ph.D., Ken Monson, Ph.D., Kirk Thibault, Ph.D., D-IBFES, Keith Button, Ph.D., PE, and Johan Ivarsson, Ph.D. (Gibbons, attorneys; Lawrence S. Lustberg and Ruth O’Herron, on the brief).

CJ Griffin submitted a brief on behalf of amici curiae Dr. Jeff Kukucka, Ph.D., Keith A. Findley, Esq., Dr. Deborah Davis, Ph.D., and Dan Simon, Esq. (Pashman Stein Walder Hayden, attorneys; CJ Griffin, Claude Caroline Heffron, Christian Martinez, and Dominique Kilmartin, on the brief).

Mary Beth Hogan submitted a brief on behalf of amici curiae Family Justice Resource Center, upEND Movement, MJCF Coalition, and Mothers Outreach Network (Debevoise & Plimpton, attorneys; Mary Beth Hogan, on the brief).

Alexandra S. Jacobs (Montgomery McCracken Walker & Rhoads) submitted a brief on behalf of amici curiae American Academy of Pediatrics, New Jersey State Chapter of the American Academy of Pediatrics, American Association for Pediatric Ophthalmology and Strabismus, American Society of Pediatric Neurosurgeons, American Society of Pediatric Neuroradiology, American Professional Society on the Abuse of Children, Society for Pediatric Radiology, and

The Ray E. Helfer Society (Montgomery McCracken Walker & Rhoads, and Kienbaum Hardy Viviano Pelton & Forrest, attorneys; Alexandra S. Jacobs, of counsel and on the brief, and David Porter (Kienbaum Hardy Viviano Pelton & Forrest) of the Michigan bar, admitted pro hac vice, on the brief).

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Richard P. Lomurro submitted a brief on behalf of amicus curiae Association of Criminal Defense Lawyers of New Jersey (Lomurro Munson, attorneys; Richard P. Lomurro, and Christina Vassiliou Harvey, of counsel and on the brief, and Andrew B. Broome, on the brief).

JUSTICE PIERRE-LOUIS delivered the opinion of the Court.

Table of Contents

I. INTRODUCTION 5

II. HISTORY OF SBS/AHT 9

III. FACTS & PROCEDURAL HISTORY 26

 A. State v. Nieves 27

 1. Facts and Pre-Hearing Proceedings 27

 2. The Frye Hearing 29

 3. Trial Court Opinion 57

 B. State v. Cifelli 61

C. Consolidated Appellate Division Decision	64
D. Review and Amicus Participation Granted	68
IV. PARTIES' ARGUMENTS	69
A. State of New Jersey and Amici Supporting the State's Position	69
B. Nieves and Cifelli and Amici Supporting Their Position	72
V. LEGAL STANDARDS	78
A. Standard of Review	78
B. Expert Testimony	78
C. Case Law Addressing SBS/AHT	83
VI. ANALYSIS	91
A. The Relevant Scientific Communities	92
B. Application of the Frye Standard of Reliability	94
VII. CONCLUSION	108

I. INTRODUCTION

Many crimes grip the hearts and minds of the public due to their cruel and heinous nature. The abuse of children, some of the most vulnerable members of our society, is certainly such a crime. Child abuse is, without question, reprehensible.

Regardless of the severity or viciousness of a crime, however, the bedrock foundations and protections embedded within our system of criminal justice must prevail. Under one of those core principles, only evidence that is sufficiently reliable and has probative value not outweighed by the evidence's

prejudicial effect may be presented to the jurors to inform their consideration of the charges against the accused.

That requirement extends to expert testimony, which must be properly admitted like all evidence. “[T]he Judiciary must ensure that proceedings are fair to both the accused and the victim. Trial judges partly fulfill that responsibility by serving as a gatekeeper.” State v. J.L.G., 234 N.J. 265, 307 (2018). Pursuant to N.J.R.E. 702, judges “must assess whether expert testimony is sufficiently reliable before it can be presented to a jury.” Id. at 308. “Given an expert witness’s singular status in the courtroom,” State v. J.R., 227 N.J. 393, 411 (2017) -- an expert witness is presented as having been deemed, by the court, sufficiently knowledgeable to offer testimony beyond the specifics of a particular case -- legislatures and courts have developed particular inquiries to assess the admissibility of proposed expert testimony.

The standard to admit expert testimony that governed at time of the proceedings in this case was the standard set forth in Frye v. United States, 293 F. 1013 (D.C. Cir. 1923).¹ Under the Frye standard, such testimony must

¹ In Olenowski I, this Court rejected the Frye standard and adopted “principles similar to the standard outlined in Daubert [v. Merrell Dow Pharms., Inc.], 509 U.S. 579 (1993)] to examine the admissibility of expert evidence in criminal and quasi-criminal cases.” State v. Olenowski (Olenowski I), 253 N.J. 133, 139, 155 (2023). Criminal matters that originated and were ruled upon before Olenowski I, such as this one, are nonetheless governed by the Frye standard

not only be reliable, but its proponent must establish that the information to which the expert will testify is generally accepted in the relevant community to which the expertise belongs.

In this appeal, we are asked to determine whether expert testimony regarding Shaken Baby Syndrome/Abusive Head Trauma (SBS/AHT)² is sufficiently reliable to go before a jury in two separate cases. In both matters, the young children exhibited symptoms that have come to be associated with SBS/AHT and referred to as the “triad of symptoms” -- subdural hematoma, retinal hemorrhages, and encephalopathy. In both cases, after the children

to assess the reliability prong. Id. at 154 (“Nothing in today’s decision disturbs prior rulings that were based on the Frye standard.”).

² Although Shaken Baby Syndrome “is a term often used by physicians and the public to describe abusive head trauma inflicted on infants and young children,” the American Academy of Pediatrics (AAP), in 2009, changed the name from Shaken Baby Syndrome to Abusive Head Trauma because “advances in the understanding of the mechanisms and clinical spectrum of injury associated with abusive head trauma compel[led the AAP] to modify [its] terminology to keep pace with . . . understanding of pathological mechanisms” of injury. Cindy W. Christian, Robert Block, & Comm. on Child Abuse & Neglect, Abusive Head Trauma in Infants and Children, 123 Pediatrics 1409, 1409 (2009). According to the AAP, it adopted the term AHT “in recognition of the fact that inflicted head injury of children can involve a variety of biomechanical forces, including shaking.” Sandeep K. Narang et al., Abusive Head Trauma in Infants and Children, 145 Pediatrics Apr. 2020, at 1, 1. The AAP noted that it “continues to embrace the ‘shaken baby syndrome’ diagnosis as a valid subset of the AHT diagnosis.” Ibid. As such, throughout this opinion we use the combined term SBS/AHT. In discussing the history of SBS/AHT, we use the terminology in use at the time.

were examined by various medical specialists and a series of tests were run, the same doctor determined that the children were victims of child abuse, specifically SBS/AHT. Because the children were in the care of their fathers when they began to exhibit the above-mentioned symptoms, both men were charged with criminal offenses, including aggravated assault and child endangerment.

The State sought to present the expert testimony of a doctor who opined, based on her findings, that the only explanation for the children's symptoms, to a reasonable degree of medical certainty, was that the children were shaken by the caregiver. That is, the anticipated testimony to the jury would be that, based on the expert's medical evaluation of the child, the child was abused.

The defense in both cases moved to exclude the doctor's expert testimony, challenging the scientific basis and reliability of the theory that shaking alone, without some other impact to the head, can cause the symptoms associated with SBS/AHT. Defendants argued that the scientific community no longer accepts that theory. After a Frye hearing, the trial court in one matter granted the defense motion and excluded SBS/AHT testimony from the trial. The second trial court followed suit. The appeals were consolidated and the Appellate Division affirmed.

For all the reasons that follow, we agree with the trial courts and Appellate Division that the State has not met its burden in establishing the reliability of SBS/AHT testimony in this case, and we affirm the Appellate Division's judgment.

II. HISTORY OF SBS/AHT

Before detailing the facts and procedural history of the matters before us, we provide the following background on the origins and evolution of SBS/AHT over the past six decades.

In 1968, neurosurgeon Dr. Ayub Ommaya conducted an experimental study focused specifically on whiplash injuries from car accidents. Although Dr. Ommaya's study did not concern head trauma in infants from shaking, it became a foundation for diagnoses of SBS/AHT, particularly its conclusion that brain injuries could occur "by rotational displacement of the head on the neck alone, without significant direct head impact." Ayub K. Ommaya et al., Whiplash Injury & Brain Damage: An Experimental Study, 204 J. Am. Med. Ass'n. 285, 285 (1968).

In the study, Dr. Ommaya subjected monkeys to a whiplash event that mimicked rear-end car collisions. Id. at 286. Specifically, anesthetized monkeys were secured to a carriage on roller-skate type wheels, equipped with a braking mechanism, that could move freely along a 20-foot-long track. Ibid.

In delivering controlled blows to the rear of the carriage, the sudden impact produced “high accelerations of the carriage and forward-facing monkey so as to mimic a rear-end collision.” Ibid. The carriage was accelerated to a speed of 30 miles per hour.³ The whiplash that resulted was recorded by “a high-speed, 16-mm movie camera at 1,000 frames/sec.” and the “rotational velocity and acceleration” of the monkeys’ heads were calculated. Ibid.

Even though the monkeys did not sustain significant direct head impact during the whiplash event, Dr. Ommaya concluded that whiplash “without direct impact to the head” can cause concussions and “produce consistent brain damage in the monkey as evidenced by subarachnoid and subdural hemorrhage [and] cerebral contusions.” Id. at 285. Dr. Ommaya’s study determined that “it is a matter of crucial importance that we investigate and manage the clinical problems of whiplash injuries” because whiplash injuries “may be of significant importance in producing the effects of closed-head injuries under conditions when the head is free to move.” Id. at 289.

³ Although the acceleration speed was not specifically noted in the 1968 study, Dr. Ommaya clarified in a later study, discussed more fully below, that the energy level of acceleration in the whiplash study “related to speeds at motor vehicle crashes at 30 mph.” A. K. Ommaya et al., Biomechanics and Neuropathology of Adult and Paediatric Head Injury, 16 Brit. J. of Neurosurgery 220, 221 (2002).

In 1971, neurosurgeon Dr. Arthur Norman Guthkelch presented the theory of SBS, partially relying on Dr. Ommaya's 1968 study. See A.N. Guthkelch, Infantile Subdural Haematoma and its Relationship to Whiplash Injuries, 2 Brit. Med. J. 430, 430 (1971). Dr. Guthkelch hypothesized that an "infant having been shaken rather than struck by its parent" might sustain a subdural hematoma. Ibid.

Dr. Guthkelch studied "[t]wenty-three cases of proved or strongly suspected parental assault on children all under the age of 3 years." Ibid. Subdural bleeding was present in 13, and in five cases no external marks of injury to the head were present. Id. at 430-31. Dr. Guthkelch concluded that "in considering any case of infantile subdural haematoma, even when there are only trivial bruises or indeed no marks of injury at all," one must inquire as to whether "the baby's head could have been shaken." Id. at 431. Dr. Guthkelch stated that the most common cause of subdural hemorrhages in infants is the rupturing of one or more of the bridging veins which run from the cerebral cortex (the outermost layer of gray matter in the brain) to the venous sinuses, id. at 430, which drain deoxygenated blood from the brain. Dr. Guthkelch posited that this rupture could occur as a result of both impact and non-impact events involving the head. Ibid.

Dr. Guthkelch did not conduct any scientific studies in support of his hypothesis, save for an experiment using a round-bottomed glass flask filled with fluid and desiccated coconut, which he both shook and struck, in order to observe the movement of the coconut in response to those actions. See id. at 431. Dr. Guthkelch analogized his observations to Dr. Ommaya's determinations regarding the movement of the brain in monkeys subjected to a whiplash event. Ibid. Dr. Guthkelch further credited Dr. Ommaya's study with giving him the idea to construct the round-bottomed flask device that informed his study and conclusions. Ibid.

In 1972, pediatric radiologist Dr. John Caffey published a paper in which he studied 27 cases believed to involve the shaking of infants or "whiplash-shaking," as he termed the action. John Caffey, On the Theory and Practice of Shaking Infants: Its Potential Residual Effects of Permanent Brain Damage and Mental Retardation, 124 Am. J. Diseases Child. 161, 161, 164 (1972). Dr. Caffey, citing to four of his own previous papers, noted that "[d]uring the last 25 years, substantial evidence, both manifest and circumstantial, has gradually accumulated which suggests that the whiplash-shaking and jerking of abused infants are common causes of" trauma to the skeleton and brain. Id. at 161.

Dr. Caffey did not conduct any biomechanical experiments himself. Instead, to arrive at his conclusions, he reviewed historical cases in which “whiplash-shaking” was suspected. See id. at 163-64. One such case -- for which he cited to an article in Newsweek -- involved a nurse who was “reported to have killed three infants and maimed 12 others . . . largely by shaking and jolting infantile brains and their blood vessels” via whiplash-shaking and “pounding on the back during burping.” Id. at 163. Dr. Caffey also cited Dr. Guthkelch’s 1971 study for examples of admitted shaking by a caregiver where subdural hematoma, retinal hemorrhages, and a broken femur were present. Ibid. Several of the cases Dr. Caffey studied involved shaking with impact, including one child’s head banging against her crib during a shaking incident and another child being “shaken and beaten to death with a stick.” Ibid.

Dr. Caffey’s study also cited cases of infant bone fractures and included numerous x-ray images of unexplained fractures of infant femurs, ankles, and arms. Id. at 162-64. In one case, an infant’s unexplained swelling and trauma to his legs was revealed to have resulted from his own brother when the parents saw the eight-year-old brother seize his infant sibling by the legs, “sh[ake] him violently, and sw[i]ng him, and fl[i]ng him onto a bed.” Id. at 164. The study does not explain whether the infant suffered cranial injuries or

subdural hematomas as a result and noted only the trauma to the infant's femurs. Id. at 163-64.

In his study, Dr. Caffey also noted that “[t]here are several apparently innocent, accepted, habitual practices, other than intentional shaking and jerking, which whiplash the head and brain, and which could lead to permanent brain damage.” Id. at 165. Dr. Caffey listed a number of activities that can result in subdural hematomas in infants, including “tossing the baby into the air,” spinning an infant around, and flipping an infant so that the infant is somersaulted forward while someone is holding the child's wrists. Ibid. The study also recommended that certain toys and equipment, such as baby bouncers; infant jumpers; seesaws; play slides; “powered cradles and powered rocking horses; trampolines; skateboards[;] and sled jumping,” should be more carefully assessed for their dangerousness or banned entirely for infants and young children. Ibid.

Dr. Caffey posited that whiplash-shaking possibly played a role in other diagnoses including “cerebral palsies and idiopathic epilepsy” and concluded that “[t]here is considerable manifest and much circumstantial evidence which indicates that whiplash-shaking and jolting of infantile heads may be major, unrecognized causes of mental retardation and permanent brain damage.” Id. at 168-69. Dr. Caffey concluded his paper by suggesting that “[t]he wide

practice of habitual whiplash-shaking for trivial reasons warrants a massive nationwide educational campaign to alert everyone responsible for the welfare of infants on its potential and actual pathogenicity.” Id. at 169. Again, Dr. Caffey did not conduct any biomechanical experiments of his own in support of that study and relied on the review of cases from other studies, including Dr. Guthkelch’s 1971 study.

In 1974, Dr. Caffey authored another study in which he coined the phrase “whiplash shaken infant syndrome.” John Caffey, The Whiplash Shaken Infant Syndrome: Manual Shaking by the Extremities with Whiplash-Induced Intracranial and Intraocular Bleedings, Linked with Residual Permanent Brain Damage and Mental Retardation, 54 Pediatrics 396, 396 (1974). In that study, Dr. Caffey reviewed purported “[d]irect evidence of trauma through admission by the parent-assailant,” but noted that although such evidence, “based on admission by the assailant, is meager, it is valuable because it is reliable.” Id. at 397. With that study, Dr. Caffey identified what he believed were the “essential elements” of whiplash shaken infant syndrome, including subdural and retinal hemorrhages, id. at 396, 402, two of the three symptoms that would later be referred to as the “triad” of symptoms indicative of shaken baby syndrome.

Dr. Caffey once again cited to Newsweek's account of the infant nurse as "[b]y far the most extensive anecdotal proof of pathogenic manual [whiplash shaking]" based on the nurse's confession to the "savage shakings of dozens of infants." Id. at 397. Dr. Caffey again referenced an eight-year-old sibling who shook his infant brother. See id. at 398. Because no citation was included for the case, it is unclear whether that is the same case he referred to in his 1972 article. But in that 1974 paper, Dr. Caffey specifically stated that subdural hematoma was not detected in that infant. See ibid.

The 1974 study also included incidents not solely related to shaking without impact, including a case of a father who first attempted to strangle his infant child and later shook the child to revive them after the infant passed out. Ibid. In that case, Dr. Caffey concluded that "it is likely that the stresses of strangling as well as those of manual [whiplash shaking] contributed to the [infant's] intraocular changes and possibly the subdural hematomas." Ibid.

Dr. Caffey further discussed the correlation between retinal hemorrhages and whiplash shaking. Id. at 400. He noted that one study found that, out of 1,238 newborns, 14% had retinal hemorrhages immediately after birth. Ibid. According to Dr. Caffey, however, that percentage dropped significantly to 2.6% between the infant's third and fifth day after birth, indicating that retinal hemorrhages due to birth disappear rather quickly, as opposed to "retinal

hemorrhages in abused infants [that can] persist[] for ten years and in one case for 19 years.” Ibid. Dr. Caffey concluded that “[t]he preponderance of the evidence from several sources indicates that the idiopathic retinal hemorrhages of the newborn infant are not due to trauma at the time of birth and the retinal hemorrhages found in battered and shaken infants are probably caused by postnatal manual shaking.” Ibid. (emphases added).

To support his whiplash shaken infant syndrome hypothesis, Dr. Caffey cited Dr. Ommaya’s 1968 whiplash study. He noted that Dr. Ommaya’s study found that concussions could occur “by rotational displacement alone of the head on the neck.” Id. at 402. Dr. Caffey posited that “[r]otation of the head is of course a consistent additional stress in the manual . . . shaking of infants.” Ibid.

In comparing the force of infant shaking with the force generated during a car accident whiplash event, Dr. Caffey stated -- without elaboration or citation -- that “[i]t is obvious that although the single manual shake of an infant may be less forceful and pathogenic than the single whiplash in an automobile accident,” manual shaking, when repeated, nevertheless “may be much more harmful to the brain” and “the veins in the eyes.” Ibid. He concluded his paper by stating that “[c]urrent evidence, though manifestly incomplete and largely circumstantial, warrants a nationwide educational

campaign” on the potential risks of whiplash shaking of infants. Id. at 403 (emphasis added). Once again, Dr. Caffey did not conduct any scientific experiments in that study and relied on anecdotal reported cases and case studies.

In the years following Dr. Caffey’s 1974 study, the theory of whiplash infant shaking syndrome, later referred to as shaken baby syndrome, began to gain traction in the medical community.⁴ That acceptance was based almost entirely on Dr. Guthkelch and Dr. Caffey’s case studies, both of which relied on Dr. Ommaya’s scientific study as the sole source of experimental scientific

⁴ There was little research regarding SBS/AHT following Dr. Caffey’s 1974 study until Dr. Duhaime’s in 1987. See Deborah Tuerkheimer, The Next Innocence Project: Shaken Baby Syndrome and the Criminal Courts, 87 Wash. U. L. Rev. 1, 12 (2009) (discussing the “new ‘evidence-based medicine’” standards that led to heightened scrutiny and a resurgence of SBS/AHT related literature “in the mid- to late-1990s”); Jack R. Shepard et al., Child Abuse and Exploitation: Investigative Techniques 26-29 (2d ed. 1992) (describing SBS/AHT); Julie Jonas, Unequal Funding Compounds Tragedy: Failures in Defending Against Shaken Baby Syndrome Charges, 96 Temp. L. Rev. 135, 148 (2024) (discussing the history of SBS/AHT’s acceptance and noting that its acceptance was “[b]ased only on the case studies by Dr. Guthkelch and Dr. Caffey” and Dr. Ommaya’s research); Christian, Block, & Comm. on Child Abuse & Neglect, 123 Pediatrics at 1409 (explaining the history of SBS/AHT’s acceptance); Wajd N. Al-Holou et al., Nonaccidental Head Injury in Children: Historical Vignette, 3 J. Neurosurgery Pediatrics 474, 480-81 (2009) (discussing Dr. Guthkelch’s and Dr. Caffey’s work as the foundation of SBS/AHT’s acceptance). But see Stephen Ludwig & Matt Warman, Shaken Baby Syndrome: A Review of 20 Cases, 13 Annals Emergency Med. 104, 104-07 (1984) (reviewing 20 incidents of SBS/AHT).

data behind the theory, notwithstanding the fact that Dr. Ommaya's study had nothing to do with the shaking of infants. See Sissoko v. State, 182 A.3d 874, 899 (Md. Ct. Spec. App. 2018) (discussing the history of SBS/AHT); see also Commonwealth v. Martin, 290 S.W.3d 59, 62-63 (Ky. Ct. App. 2008) (describing testimony that discussed the history of SBS/AHT).

In 1987, Dr. Ann-Christine Duhaime conducted the first biomechanical study testing shaken baby syndrome and the effects of shaking without impact. Ann-Christine Duhaime et al., The Shaken Baby Syndrome: A Clinical, Pathological, & Biomechanical Study, 66 J. Neurosurgery 409, 409 (1987). The study authors reviewed all cases of SBS at the Children's Hospital of Philadelphia (CHOP) from January 1978 to March 1985. Id. at 409-10. Out of the 48 cases for which clinical data was available, only one involved shaking without impact, and 15 cases -- the largest percentage -- were cases of accidental blunt trauma, usually from a fall. Id. at 410. After reviewing the cases, Dr. Duhaime determined that pathological examinations revealed that "all of the children who died had evidence of blunt head trauma." Id. at 411. Those findings of impact, however, were visible only at the time of autopsy in seven of the thirteen cases resulting in death because the trauma was not apparent prior to the children's deaths. Ibid.

Importantly, after conducting a biomechanical study using infant models, Dr. Duhaime concluded that “shaking alone does not produce the shaken baby syndrome.” Id. at 409. Dr. Duhaime used infant models and implanted “an accelerometer to measure the results of shaking or impact manipulations.” Id. at 411. Dr. Duhaime sought to determine what level of force, both acceleration and deceleration, is generated by shaking alone and, separately, by impact to the head. See id. at 413. The models were shaken violently and repetitively. Ibid. The back of models’ heads were then struck against a metal bar or a padded surface. Ibid. According to Dr. Duhaime, the results showed that the acceleration and velocity levels for shaking fell well below the threshold at which injuries, such as concussions and subdural hematomas, occurred in previous studies using primates. Id. at 414. The results further showed that the forces generated as a result of impact fell within the acceleration and velocity ranges expected in cases in which concussions, subdural hematomas, and diffuse axonal injury, a type of traumatic brain injury, were present.⁵ Ibid. In short, Dr. Duhaime’s study was not able to

⁵ Specifically, Dr. Duhaime determined that the average peak acceleration for the shaking episodes was 9.29 g, whereas the average peak acceleration for the impact events was 428.18 g, over 50 times more. Id. at 413. A “g” is the “force of gravity or acceleration on the body.” Merriam-Webster Dictionary, <https://www.merriam-webster.com/dictionary/g-force> (last visited Oct 31, 2025).

generate, in the infant models, the forces necessary to result in the relevant brain injuries by shaking alone. See id. at 413-14.

Based on her biomechanical experiment and review of SBS cases at CHOP, Dr. Duhaime concluded that “shaken baby syndrome, at least in its most severe acute form, is not usually caused by shaking alone.” Id. at 414. Dr. Duhaime added that “[a]lthough shaking may, in fact, be a part of the process, it is more likely that such infants suffer blunt impact. The most common scenario may be a child who is shaken, then thrown into or against a crib or other surface, striking the back of the head.” Ibid. (emphasis added).

In 1993, the American Academy of Pediatrics (AAP) released its first statement on the theory of SBS. Citing in part to both Drs. Guthkelch and Caffey, the AAP’s statement noted that “data regarding the nature and frequency of head trauma consistently support a medical presumption of child abuse when a child younger than 1 year of age has intracranial injury.” Am. Acad. of Pediatrics, Comm. on Child Abuse & Neglect, Shaken Baby Syndrome: Inflicted Cerebral Trauma, 92 Pediatrics 872, 872 (1993) (emphasis added). The AAP’s statement mentioned Dr. Duhaime’s experimental study, simply noting that “a challenge to the presumption that the shaking alone is the sole source of the trauma” arose from Dr. Duhaime’s study. Ibid. The AAP statement related to children believed to have been

shaken, with or without impact. Ibid. The statement advised medical professionals on the clinical features of SBS and the role of specialized disciplines in recognizing SBS. Id. at 873-74. The statement further suggested that a diagnostic team of specialists be formed, with medical professionals in pediatrics, radiology, neurology, neurosurgery, and ophthalmology. Id. at 874. The AAP's statement did not discuss the Guthkelch and Caffey studies in depth and did not mention or cite to Dr. Ommaya's 1968 study.

In 2002, Dr. Ommaya, whose whiplash study was the basis for Dr. Guthkelch's conclusions and Dr. Caffey's whiplash shaken infant syndrome hypothesis, published a study criticizing other researchers' reliance on his own study as the scientific foundation of SBS. A.K. Ommaya et al., Biomechanics and Neuropathology of Adult and Paediatric Head Injury, 16 Brit. J. Neurosurgery 220 (2002).

Dr. Ommaya cautioned that "[i]t is improbable that the high speed and severity of the single whiplash produced in our animal model could be achieved by a single manual shake or even a short series of manual shaking of an infant in one episode." Id. at 221. Dr. Ommaya further noted that Dr. Caffey, Dr. Guthkelch, and others referenced by analogy his 1968 study "not realizing that the energy level of acceleration in [his] work related to speeds at

motor vehicle crashes at 30 mph.” Ibid. Dr. Ommaya acknowledged that “it is possible that continued repetitive shaking over a period of time, or shaking repeated at intervals can produce significant cerebral as well as cervical spinal cord trauma in an infant.” Id. at 225 (emphasis added). Regarding the biomechanics of retinal hemorrhages, Dr. Ommaya opined that “the levels of force required for retinal bleeding by shaking to damage the eye directly is biomechanically improbable,” given that small masses like the eye require levels of force even higher than larger masses like the brain. Id. at 233.

Significantly, Dr. Ommaya critiqued the assumptions upon which theories about mechanisms of pediatric retinal and brain injuries in literature have been based. Id. at 227. Those assumptions included the following: short falls cannot cause subdural hematomas; retinal hemorrhages result from shaking; and the time interval between the onset of symptoms as a result of SBS is always brief. Ibid. Dr. Ommaya stated that “[t]hese assumptions individually and in concert are ambiguous or incorrect” even though those very assumptions had been used as the bases for the differential diagnoses of SBS, “usually without reference to available biomechanical analysis.” Ibid.

A decade after Dr. Ommaya signaled that SBS lacked an underlying scientific foundation, Dr. Guthkelch, whose study relied on Dr. Ommaya’s whiplash experiment and was in turn cited by Dr. Caffey, also questioned the

science behind SBS. A.N. Guthkelch, Problems of Infant Retino-Dural Hemorrhage with Minimal External Injury, 12 Hous. J. Health L. & Pol’y 201, 201 (2012). In his article, Dr. Guthkelch noted that the controversy surrounding SBS/AHT had risen to a level of emotion and divisiveness “that has interfered with our commitment to pursue the truth.” Ibid.

According to Dr. Guthkelch, his 2012 article, which was not a study, was “a call for civility in scientific discourse.” Ibid. Dr. Guthkelch identified several problems with the SBS/AHT theory and concluded that given the science, it does not follow that “one can infer shaking (or any other form of abuse) from a finding of retino-dural hemorrhage in infancy.” Id. at 203. Dr. Guthkelch pointed out that there “seem to have been instances in which both medical science and the law have gone too far in hypothesizing and criminalizing alleged acts of violence” when the only evidence was the presence of the triad of symptoms or just one or two symptoms of the triad. Id. at 203-04.

Dr. Guthkelch further stated that “SBS and AHT are hypotheses that have been advanced to explain findings that are not yet fully understood.” Id. at 207. In acknowledging that there is nothing wrong with advancing hypotheses, Dr. Guthkelch noted that “[i]t is wrong, however, to fail to advise parents and courts when these are simply hypotheses, not proven medical or

scientific facts, or to attack those who point out problems with these hypotheses or who advance alternatives. Often, ‘getting it right’ simply means saying, clearly and unequivocally, ‘we don’t know.’” Ibid.

In 2018, while the SBS/AHT debate continued, a group of physicians and pediatric radiologists published a consensus statement on AHT (the Consensus Statement) which the AAP later endorsed. The Consensus Statement purports to “address[] significant misconceptions and misrepresentations about the diagnosis of [AHT] in infants and young children,” noting that, “[r]ecently, denialism of child abuse has become a significant medical, legal and public health problem.” Arabinda Kumar Choudhary et al., Consensus Statement on Abusive Head Trauma in Infants and Young Children, 48 Pediatric Radiology 1048, 1050 (2018). The Consensus Statement further noted that “the courtroom has become a forum for speculative theories” regarding SBS/AHT, despite the fact, according to the Consensus Statement, that “[SBS/]AHT is a scientifically non-controversial medical diagnosis.” Id. at 1048-49. As a result of what the authors deemed SBS/AHT “denialists” and the sensationalized critique of SBS/AHT in the media, the Consensus Statement authors stated that they hoped the document would reduce confusion and help courts “recognize unsubstantiated medical expert testimony.” Id. at 1049, 1060.

The Consensus Statement cited to the studies by Dr. Caffey and noted Dr. Caffey's citation to Dr. Ommaya and Dr. Guthkelch. Id. at 1051. Among other conclusions, the authors of the Consensus Statement discounted short-distance falls as a possible cause of subdural hematomas and retinal hemorrhages. Id. at 1052. They acknowledged Dr. Duhaime's conclusion that shaking alone cannot generate the force needed to cause the injuries that result in SBS/AHT cases but noted that contrary evidence -- confessions by caregivers -- supports the argument that shaking alone can cause SBS/AHT. Id. at 1051. The Consensus Statement also noted that SBS/AHT is a medical diagnosis, not a legal finding of murder, contrary to the arguments of defense attorneys. Id. at 1059. The Consensus Statement declared quite definitively that "[t]here is no controversy concerning the medical validity of the existence of AHT." Id. at 1048.

III. FACTS & PROCEDURAL HISTORY

We next turn to the facts and procedural history of the two cases that gave rise to this appeal.

A. State v. Nieves⁶

1. Facts and Pre-Hearing Proceedings

Respondent Darryl Nieves is the father of D.J., who was born prematurely at 25 weeks due to complications his mother experienced during her pregnancy. After his birth in March 2016, D.J. remained at Saint Peter's University Hospital through October 2016, aside from two hospital stays at CHOP for cardiac surgery in May and July 2016. After being discharged, D.J. resided with his mother and Nieves, who were his primary caregivers.

In February 2017, D.J. -- 11 months old at the time -- had three episodes of seizure-like behavior over the course of two weeks. During the first episode, Nieves was home alone with D.J. and was changing his diaper when D.J. went limp and passed out. Nieves attempted to revive D.J. by blowing in his mouth, and he called D.J.'s mother and 9-1-1. By the time paramedics arrived, D.J. was alert. The parents took D.J. to the pediatrician to follow up and he was diagnosed with acid reflux. A few days later, the second seizure episode occurred. While Nieves was changing D.J.'s diaper, the child passed out and went limp again. Nieves applied oxygen with a nasal cannula, a device that delivers supplementary oxygen to someone in need of respiratory

⁶ The facts in this section are derived from D.J.'s medical records and testimony elicited at the Frye hearing.

assistance. The third episode began several days later. Nieves picked up D.J. to change his diaper and D.J. went stiff and his jaw locked. Nieves brought D.J. upstairs to his mother, and they called an ambulance. She took a video of D.J. that showed him not responding to stimulation, with one arm stiff and the other flexed, and eyes deviated to the left. The ambulance transported D.J. to Saint Peter's where he was admitted.

Those events triggered a child abuse investigation. Dr. Gladibel Medina, a child abuse pediatrician and the medical director at the Dorothy B. Hersch Regional Child Protection Center at Saint Peter's, prepared a report after reviewing D.J.'s medical history and interviewing Nieves and D.J.'s mother. While at the hospital, medical staff discovered that D.J. had bilateral retinal hemorrhages -- described as extensive, too numerous to count, involving multiple layers of the retina and extending to the periphery -- and bilateral subdural bleeding with both new and old blood present. Dr. Medina noted in her report that D.J. "had a detailed review of his neonatal course and comprehensive medical workup and follow-up by pediatricians and pediatric subspecialists," which included doctors with expertise in neurology, genetics, hematology, neuroradiology, and pediatric ophthalmology, as well as a retinal specialist.

That comprehensive medical evaluation did not reveal any pathology to account for D.J.'s bilateral subdural bleeds, extensive bilateral retinal hemorrhages, or neurological symptoms observed in early February 2017. Dr. Medina noted in her report that D.J. did not sustain any reported accidental injuries in February 2017. Thus, based on D.J.'s history and symptoms when he was admitted, Dr. Medina diagnosed D.J. with SBS/AHT, with or without impact, "within a reasonable degree of medical certainty."

In the wake of Dr. Medina's diagnosis, on June 30, 2017, a Middlesex County grand jury indicted Nieves, charging him with second-degree aggravated assault and second-degree endangering the welfare of a child. Judge Pedro J. Jimenez, Jr., presided over the matter.

2. The Frye Hearing

In July 2018, Nieves moved before the trial court for a Frye hearing to challenge the scientific reliability of the SBS/AHT hypothesis and to preclude Dr. Medina's SBS/AHT testimony at trial. Nieves argued that SBS/AHT was no longer accepted in the scientific community. The court initially granted Nieves's request for a Frye hearing. Thereafter, the State filed a motion for reconsideration. Based in part on Judge Benjamin S. Bucca, Jr.'s denial of a Frye hearing in State v. Cifelli, which we discuss later, the court granted the State's motion and reversed its prior order, effectively denying Nieves a Frye

hearing. The Appellate Division granted Nieves's motion for leave to appeal and remanded for a Frye hearing. Judge Jimenez conducted that hearing over the course of five days.

The State presented one witness, Dr. Medina. Nieves presented three expert witnesses: (1) Dr. Joseph Scheller, who testified as an expert in the fields of pediatric neurology and neuroimaging; (2) Dr. Julie Mack, who testified as an expert in the fields of radiology and pediatric radiology; and (3) Dr. Chris Alan Van Ee, who testified as an expert in biomechanics. Additionally, the parties introduced into evidence and discussed numerous scientific studies and articles during the hearing.

a. Dr. Gladibel Medina

According to Dr. Medina, the evaluation of a child when SBS/AHT is suspected involves consultation with specialists including geneticists, hematologists, radiologists, and ophthalmologists to determine whether there is a possible disease, medical issue, or pathology that might be contributing to the child's symptoms. Dr. Medina testified that such specialists work together to provide child abuse pediatricians with a full history of the child's health. Dr. Medina testified that the SBS/AHT diagnosis is widely accepted within the medical community and accepted by all the pediatric subspecialties involving intracranial injury. On cross-examination, Dr. Medina agreed that there are no

specific diagnostic criteria to define SBS/AHT; there are only symptoms a child may exhibit when doctors are looking to see whether to diagnose SBS/AHT.

Dr. Medina testified about the history and origins of SBS/AHT. Dr. Medina stated that “the field of abusive head trauma or the recognition of inflicted head injury in medicine is about 160 years” old. Dr. Medina discussed medical literature from the mid-19th century that identified injuries in children believed to be “associated with maltreatment by care givers.” Dr. Medina then noted that approximately 80 years later, Dr. Guthkelch identified subdural hematomas in children with no external signs of trauma -- which was strongly associated with physical abuse. She further testified that in the 1970s, Dr. Caffey began using terminology that referred to inflicted trauma in infants caused by shaking-type injury.

Dr. Medina also testified about the study Dr. Ommaya conducted in 1968. She explained that “what we know about shaking, and the established thresholds for intracranial injury comes from that study” and that “everything else in biomechanics is based on those injury thresholds” from that study. Dr. Medina stated that the validity of SBS/AHT as a diagnosis has not changed in the medical community but has been challenged in terms of the mechanism of shaking as the cause of injury in the scientific community studying

biomechanics. She submitted that the controversy about SBS/AHT in biomechanics is focused on whether shaking can cause the forces needed to generate intracranial injury in infants.

Regarding other biomechanical studies inspired by Dr. Ommaya, Dr. Medina acknowledged that Dr. Duhaime's study found that shaking alone did not generate enough rotational forces but shaking with impact did. Dr. Medina agreed that, since Dr. Duhaime's 1987 study, there is debate about whether shaking alone can reach the force that would cause retinal hemorrhages and subdural hematomas. Dr. Medina explained that the Duhaime study was confirmed by Dr. Michael T. Prange,⁷ who, in 2003, used a different wooden mass body-type surrogate to conclude that the threshold for intracranial injury by shaking must produce force like that involved in a short (or higher) distance fall, and that one could not reach the minimum established threshold by shaking alone.

Regarding biomechanics, when defense counsel asked, "when we talk about biomechanics, your opinion is based on acceleration and deceleration of a baby," Dr. Medina answered affirmatively, and added "[o]f the baby's head." When asked whether she could explain acceleration and deceleration forces

⁷ Michael T. Prange et al., Anthropomorphic Simulations of Falls, Shakes, and Inflicted Impacts in Infants, 99 J. Neurosurgery 143 (2003).

beyond her answer that “[i]t’s just movement of the head in different planes inside the intracranial cavity,” Dr. Medina stated that she could not.

Dr. Medina further testified about two 2016 studies -- one conducted by Carole A. Jenny⁸ and another conducted by C.Z. Cory⁹ -- that used different models and changed the pattern of the shaking. Dr. Medina testified that the researchers varied the biomechanics of the dolls, which allowed for chin-to-chest impact and impact between the very back of the skull and the back. Dr. Medina noted that those studies, which as she testified included shaking with some impact, “actually surpassed the injury thresholds produced by the original Ommaya study.”

When asked about studies conducted by Dr. John W. Finnie, in which anesthetized lambs were shaken, Dr. Medina agreed that all the lambs had spinal injuries and two out of seven in the first study had retinal hemorrhages. In Finnie’s 2010 and 2012 studies, the lambs were vigorously shaken 10 times for 30 seconds over the course of 30 minutes without head impact to determine what injuries would occur with shaking alone. J.W. Finnie et al.,

⁸ Carole A. Jenny et al., Biomechanical Response of the Infant Head to Shaking: An Experimental Investigation, 34 J. Neurotrauma 1579 (2017).

⁹ C.Z. Cory & M.D. Jones, Can Shaking Alone Cause Fatal Brain Injury? A Biomechanical Assessment of the Duhaime Shaken Baby Syndrome Model, 43 Med. Sci. L. 317 (2003).

Neuropathological Changes in a Lamb Model of Non-accidental Head Injury (the Shaken Baby Syndrome), 19 J. Clinical Neurosci. 1159 (2012); John W. Finnie et al., Diffuse Neuronal Perikaryal Amyloid Precursor Protein Immunoreactivity in an Ovine Model of Non-accidental Head Injury (the Shaken Baby Syndrome), 17 J. Clinical Neurosci. 237 (2010). The 2010 study noted that a “small subdural haemorrhage was found in two shaken lambs” out of the seven lambs tested, and “retinal haemorrhages were minimal and only seen in two” lambs. Finnie et al., 17 J. Clinical Neurosci. at 239. In the 2012 Finnie study that used nine lambs of varying weights, the study noted that three of the nine lambs, the lower-weight lambs, died before the six-hour mark at which the researchers planned to kill and then study the lambs. Finnie et al., 19 J. Clinical Neurosci. at 1160. Although the 2012 study mentions that “retinal damage” was found in the shaken lambs, “retinal hemorrhages” were not specifically noted as they were in the 2010 study, and in 2012 the authors stated that “no [retinal] haemorrhage was found in the serial histological sections of the retina.” Id. at 1159, 1162-64. Dr. Finnie’s 2012 study further

noted observed injury to the brainstem¹⁰ and craniocervical junction.¹¹ Id. at 1159, 1161. Although the authors of Finnie’s 2012 study concluded that shaking alone resulted in the death of the three lower-weight lambs and that the lack of evidence of impact does not negate the occurrence of non-accidental head injury if the head is impacted on a soft surface, the authors nevertheless advised that a “diagnosis of [non-accidental head injury] should be concluded with caution, unless there is other corroborating evidence of abuse or a convincing admission by the perpetrator.”¹² Id. at 1164.

¹⁰ The brainstem is “[t]he part of the brain that is connected to the spinal cord. The brain stem is in the lowest part of the brain (just above the back of the neck) and is made up of the midbrain, pons, and medulla oblongata.” Nat’l Cancer Inst., Brain Stem, <https://www.cancer.gov/publications/dictionaries/cancer-terms/def/brain-stem> (last visited Oct. 30, 2025).

¹¹ “The craniocervical junction forms the bridge between the skull and the spine, a highly mobile group of joints that allows the mobility of the head in every direction. [It] plays a major role in protecting the inferior brainstem and spinal cord.” Juliette Raoul-Duval et al., Geometric Growth of the Normal Human Craniocervical Junction from 0 to 18 Years Old, 245 J. Anatomy 842 (2024) (parentheticals omitted).

¹² Finnie’s lamb studies suffer from the same limitations of other studies in attempting to replicate an infant model. Finnie himself recently acknowledged the difficulties in the use of animal and mechanical models in noting that “[d]ue largely to irreconcilable anatomic species differences between animal brains and human infants, and a lack of resemblance of the shaking induced by mechanical devices to real-world human neurotrauma, no animal model has been able to reliably reproduce the full range of neuropathologic AHT changes.” John W. Finnie & Peter C. Blumbergs, Animal Models of Pediatric Abusive Head Trauma, 38 Child’s Nervous Sys. 2317 (2022). Finnie further

Dr. Medina also discussed benign enlargement of the subarachnoid space (BESS). She described BESS as a medical diagnosis that puts infants at an increased risk for subdural trauma, which she described as trauma to the bridging veins that come from the brain to the dura (the sinus drainage). Dr. Medina explained that BESS is associated with trauma to the bridging veins with minimal movement involved, which causes subdural bleeding, predisposes the infant to trauma, and is called benign because it does not cause outward signs of trauma. Dr. Medina stated that to differentiate between BESS and SBS/AHT, doctors must look at the whole clinical picture and should never make a diagnosis of SBS/AHT based on one finding.

Dr. Medina noted that doctors rely on the “triad” of symptoms to flag concern for SBS/AHT and to conduct further investigation. She stated that the triad of symptoms presents as the following: (1) subdural hemorrhages (bleeding in the brain); (2) severe retinal hemorrhages (bleeding in the retina/eye); and (3) any neurological presentation known as encephalopathy, which presents as unresponsiveness, apnea, seizures, or altered mental states.

stated that “it is unlikely that any animal model will be able to precisely replicate the complete range of brain and ocular lesions to support a diagnosis of AHT in human infants.” Id. at 2322.

She explained that there are different types of subdural hemorrhages based on the cause of the hemorrhage. Most of the time, she stated, subdural hemorrhages are caused by trauma; among young children, the most common trauma is birth, and birth-related subdural hemorrhages usually resolve four to six weeks after birth.

Dr. Medina testified that subdural hematomas can be associated with retinal hemorrhages and stated that those observed in inflicted injury cases have a distinct pattern that differs from hemorrhages caused by disease, illness, or accidental trauma aside from motor vehicle roll-overs or other certain medical conditions. She testified that when subdural hematomas coexist or are identified in a child with severe retinal hemorrhages -- meaning hemorrhages that are multilayered and too numerous to count -- it raises even more of a concern for SBS/AHT.

According to Dr. Medina, if the symptoms of the triad remain unexplained after a thorough review, then medical literature confirms the symptoms “to be more specific for inflicted head injury.” Dr. Medina testified that the SBS/AHT diagnosis thus requires an elimination of other possible causes of the infant’s symptoms. Dr. Medina stated that the diagnosis is a multidisciplinary process and noted that the diagnostic process and diagnosis

itself are well-established and widely accepted in the medical community, and are reliable.

Regarding D.J., the child in Nieves, Dr. Medina testified that she evaluated him on February 15, 2017, five days after he was admitted to Saint Peter's. Dr. Medina stated that after D.J. arrived at the hospital, doctors evaluated him, looking for any other potential abnormalities. D.J. had an ophthalmological examination that revealed severe multilayered retinal hemorrhages in both eyes. Following that examination, the hospital contacted the Division of Child Protection and Permanency (DCPP), and enlisted the assistance of the Dorothy B. Hersh Child Protection Center in its evaluation of D.J. Dr. Medina explained that a neuroradiologist diagnosed D.J. with subdural bleeds and a pediatric ophthalmologist diagnosed him with severe retinal hemorrhages, which the ophthalmologist documented was "consistent with nonaccidental trauma."

Dr. Medina confirmed that after obtaining the findings of the subspecialists, she met with D.J.'s parents, who explained that between February 3 and February 10, 2017, D.J. experienced three different episodes while Nieves was the primary caregiver. Noting that D.J. had heart defects, Dr. Medina described D.J.'s birth history as very complicated because he was born extremely premature. She testified that D.J. had mild retinopathy, or

abnormally growing blood vessels, in the back of his eye when he was born, but he was reevaluated at six months and was found to have healthy mature retinas without any abnormalities. Regarding his altered mental state, Dr. Medina noted that D.J. was a healthy baby and that there were no concerns in terms of seizure-like activity until the events in February 2017. On cross-examination, however, Dr. Medina conceded that when a baby is born at 25 weeks as D.J. was, there would be medical problems present at birth. She further stated that although the cause is unknown, male premature babies are especially prone to subdural hematomas at birth and that “there are problems that can present months later.”

Dr. Medina found that D.J., then 11 months old, was developmentally delayed, was at the developmental stage of a three or four-month-old infant, and was starting to roll over but could not do much else. Though D.J. was able to babble and smile, he did not have good head control and had decreased muscle tone for his age, which, Dr. Medina explained, is not unusual when it comes to children born as prematurely as D.J.

Dr. Medina stated that D.J.’s parents denied any history of accidental trauma. Both parents mentioned that D.J.’s half-brother had been jumping in the crib with him the month prior, a few days before the first incident, but the parents reported that D.J. was smiling at the time and did not appear injured.

Dr. Medina stated that she diagnosed D.J. with “abusive head trauma through shaking” within a reasonable degree of medical certainty. Dr. Medina testified that she reached that diagnosis because D.J. presented to the hospital with “altered mental status, subdural hemorrhages, and retinal hemorrhages in the pattern that is severe and usually associated with very specific circumstances,” but did not have evidence of hyperacute increase in intracranial pressure, an aneurysm, or potential accidental trauma. Dr. Medina testified that D.J.’s symptoms were not accounted for by a metabolic disorder or accident; and every other possible explanation was ruled out by the treating providers.

Regarding her report, in which she wrote that the tearing of the bridging vein in D.J.’s brain caused his subdural hematomas, Dr. Medina conceded that no studies show a relation between tearing of a bridging vein and subdural hematomas.¹³ Dr. Medina further testified that she made an assumption in coming to that conclusion. Dr. Medina confirmed that her physical

¹³ The dissent appears to argue that Dr. Medina’s concession that she is unaware of any studies that show a relation between the tearing of the bridging vein and subdural hematomas is of no consequence and cites the Consensus Statement to support that argument. Post at ____ (slip op. at 11 n.3.) To the contrary, the fact that the State’s expert witness is unaware of purportedly well-documented information regarding a factor considered in the medical diagnosis she made in this case supports the unreliability of her testimony, particularly since Dr. Medina wrote in her report that D.J.’s subdural hematoma was caused by the tearing of the bridging vein.

examination of D.J. did not reveal any bruises on his arms, neck, or rib cage, or any spinal injuries. Although Dr. Medina's report stated a diagnosis of "abusive head trauma, as it occurs with a shaking event with or without impact," during her testimony on cross-examination, Dr. Medina agreed that "with impact" would mean "D.J. was shaken and hit against something," and Dr. Medina confirmed that there was no indication that D.J.'s head hit against anything.¹⁴

The trial court also asked Dr. Medina some clarifying questions.

[The court]: These diagnoses that you make to conclude [SBS/AHT], you used the word probabilities. Is that really all they're pretty much based on? An elimination of factors, and what's left is a probable result?

[Dr. Medina]: Elimination of things that can account for the findings.

. . . .

[The court]: [W]e can agree that we're talking about possibilities, or probabilities[?]

[Dr. Medina]: Yes.

[The court]: Not anything certain[], okay? And you reached these probabilities and possibilities by way of process of elimination[?]

¹⁴ Dr. Medina later testified in response to questions from the trial court that D.J. did not "have any external signs of impact."

[Dr. Medina]: And what's been documented in the literature.

In further response to the trial judge's questions, Dr. Medina explained that there was no way to specifically test for the exact cause of D.J.'s retinal hemorrhages. However, she stated that the patterns of retinal hemorrhages he presented are known to be associated with inflicted and traumatic injuries, and that the cases involving confessions confirmed this. The court then asked, "even the literature reaches all conclusions by way of process of elimination, right? Based on testing done?" Dr. Medina answered, "[b]ased on what is seen with accidents and not." The court asked Dr. Medina whether "the best [she] can do is say that there was some kind of inflicted trauma?," to which she responded "yes."

b. Dr. Joseph Scheller

Dr. Joseph Scheller is a pediatric neurologist currently employed in private practice who testified for the defense. Dr. Scheller noted that he has been involved with studying and reviewing SBS/AHT for approximately 20 years and that he is aware of the modern debate about SBS/AHT.

Regarding biomechanics, Dr. Scheller noted that biomechanical studies are ways to measure forces. He stated that the first biomechanical study of shaking alone was conducted by Dr. Duhaime in 1987. He stated that Dr.

Duhaime's experiment could not create forces inside the head that were powerful enough to create a subdural hematoma. On cross-examination, Dr. Scheller agreed that Dr. Duhaime's ultimate conclusion was only that SBS/AHT in its most severe acute form, meaning fatality, could not usually be caused by shaking alone. Dr. Scheller further testified that there have not been any subsequent biomechanical studies that have been able to produce the amount of force needed to cause a baby to suffer a subdural hematoma with shaking alone. Dr. Scheller acknowledged that he did not know what happens when an adult shakes an infant six-months or younger but noted that he was sure that violent shaking is "not a good thing." Dr. Scheller further noted that nobody knows for sure what happens because there have been no reports of witnessed shaking where the result was subdural hematomas, retinal hemorrhages, or neck injury.¹⁵ He explained that for almost any other disease

¹⁵ In their briefing, both the State and the defense cite to a study involving witnessed shaking that was published after Dr. Scheller testified at the Frye hearing. See Kenneth W. Feldman et al., Abusive Head Trauma Follows Witnessed Infant Shaking, 31 Child Abuse Rev., 2022, at 1, 1. The State argues that the Feldman study supports the theory that shaking alone causes the symptoms associated with SBS/AHT. The defense argues that the study's results do not support that proposition given the small percentage of children studied that experienced symptoms associated with SBS/AHT and the limitations of the study. In the Feldman study, researchers examined 23 cases of witnessed shaking without impact and described 10 in greater detail. Id. at 3. Of those 23 cases, five children experienced subdural hemorrhages, retinal hemorrhages, and neurological symptoms; another five experienced

or medical problem, there is quite good scientific data, but for SBS/AHT, the scientific data is sorely lacking.

Regarding confessions, Dr. Scheller contrasted confessions, which he characterized as unreliable due to coercion and inaccuracies, and medical history, freely provided by a patient or caregiver to a doctor when asked why they are seeing a doctor.

In terms of specific symptoms and their relationship to SBS/AHT, Dr. Scheller explained that although subdural hematomas and retinal hemorrhages can be symptoms of SBS/AHT, a child can be diagnosed with SBS/AHT absent those symptoms. Further, he noted that the same was true of seizures, bruising, broken ribs or bones, neck injuries, or other external signs of trauma.

Dr. Scheller then described the report he wrote regarding D.J. He stated that he reviewed D.J.'s birth and nursery records, pediatric visits, his Saint Peter's hospitalization in 2017, and all of the radiology images -- X-rays,

neurological symptoms but no subdural or retinal hemorrhaging. Id. at 3-6. Of the five children that experienced the three symptoms associated with SBS/AHT, two had other signs of abuse, including multiple femur and tibia fractures as well as healing rib fractures. Id. at 6. The study noted that "[a]lthough shaking is rarely witnessed, these cases support that shaking alone can cause typical AHT injuries." Id. at 1. The authors, however, identified limitations of the study in noting that "[w]itness statements might be inaccurate or modified by self-interest. Problems with witness observations, recollection and completeness of their statements could hamper some results." Id. at 8.

ultrasound scans, CAT scans, and MRI scans. In his report, Dr. Scheller constructed a timeline of what happened medically to D.J. and concluded that there was no evidence at all that D.J. was a victim of abuse.

Dr. Scheller then testified about D.J.'s MRI images taken in February 2017. In certain images he pointed out what he described as a subdural hygroma, which is fluid that collects in the space between the brain and the skull that the body does not know how to get rid of. He testified that a subdural hygroma is caused by a minor trauma. He noted that subdural hygromas are a common finding in children born prematurely.

Dr. Scheller stated that he believed D.J.'s subdural hygroma was caused by his prematurity. He reminded the trial court that D.J. was born extremely prematurely and weighed only 600 grams (approximately 1.3 pounds) at birth. He explained that sometimes there can be no symptoms associated with hygromas and at other times there can be poor feeding and poor weight gain because there is pressure from that fluid on the brain, which causes irritation and can cause seizures and sometimes delays in development.

Dr. Scheller explained that a bridging vein is a big vein usually found near the top of the brain that collects blood and brings it back to the larger blood vessels that deliver it to the heart. He stated that if a bridging vein ruptured, it would cause a large blood clot. Dr. Scheller testified that a blood

clot found in D.J.'s MRI was not large and was just a sliver of a blood clot. Furthermore, he believed D.J. had a subdural hygroma and that he had a seizure on or around February 10 that certainly could have been caused by the large fluid collection that was irritating his brain. He explained that such a fluid collection could cause neurological symptoms because it squeezes the brain a little and the brain can become irritated.

Defense counsel asked Dr. Scheller about what conditions were known to cause retinal hemorrhages, to which he replied that the number one condition in the world that causes them is "being born normal." He explained that one-fifth to one-third of perfectly normal babies in the nursery have retinal hemorrhages. He then testified regarding a study performed by Dr. Natalia Callaway¹⁶ on retinal and optic nerve hemorrhages in newborn infants in 2016. He explained that in the study, researchers looked at more than 200 perfectly normal newborns shortly after they were born to examine their eyes and determine if they had retinal hemorrhages. He stated that researchers found that 20 percent had retinal hemorrhages and that out of that 20 percent, about 70 percent had multilayer retinal hemorrhages. Dr. Scheller further expressed on cross-examination that pediatric ophthalmologists are experts in describing,

¹⁶ Natalia F. Calloway et al., Retinal and Optic Nerve Hemorrhages in the Newborn Infant: One-Year Results of the Newborn Eye Screen Test Study, 123 Ophthalmology 1043 (2016).

detailing, and making sure that retinal hemorrhages do not affect the vision and can perform surgery, if needed, to repair the eye, but they are not experts in determining the cause of retinal hemorrhages.

Dr. Scheller testified that babies develop retinal hemorrhages at birth because their heads are squeezed during the birthing process, creating circulation problems in the brain, and because of the dramatic change from fetal circulation to normal circulation. Dr. Scheller opined that he believed D.J.'s retinal hemorrhages were caused by an accumulation of too much fluid and too much pressure in between the brain and the inside of his skull. He explained that children who are abused often have indicia such as fractured bones, neck injury, bruises, or internal organ injuries, but D.J. did not. He stated that he believed D.J. has another condition, hygroma, that mimics SBS/AHT that Dr. Medina did not consider. Dr. Scheller noted that nowhere in Dr. Medina's report did she rule out chronic hygroma as a diagnosis. Dr. Scheller also said that he did not rule out SBS/AHT in D.J.'s case, but he noted that there are no specific criteria in the diagnosis of SBS/AHT, and that there were very few findings consistent with an SBS/AHT diagnosis in D.J.'s case. Dr. Scheller confirmed that in his medical opinion, to a reasonable degree of scientific certainty, D.J. was not the victim of SBS/AHT.

On redirect, defense counsel asked Dr. Scheller one question: whether there was “a study that shows a human can shake a baby causing the triad of injuries,” to which Dr. Scheller responded that “[t]here is not.”

c. Dr. Julie Mack

Dr. Julie Mack, M.D., an assistant professor of radiology at the Penn State College of Medicine at the Penn State Milton S. Hershey Medical Center in Hershey, Pennsylvania, was qualified as an expert in the field of radiology and pediatric radiology.

Dr. Mack testified about subdural collections around the brain and stated that such collections can occur without trauma and without predisposing conditions. Dr. Mack stated that the term subdural is used to refer to anything that disrupts the connection between the arachnoid and the dura -- it can be fluid, old blood, new blood, or a combination of blood plus fluid. She explained that sometimes subdural hemorrhage is used as a catch-all phrase.

Next, Dr. Mack explained that BESS is an anatomic variation that is poorly understood and occurs when there is more fluid than is typically seen around the brain -- sometimes in the subarachnoid space, sometimes in the subdural space. She stated that some infants who have BESS may have seizures, or changes in the eyes such as a downward gaze and lethargy. Dr. Mack further noted that it is unknown how many patients have BESS but do

not present with symptoms. Dr. Mack agreed with counsel that an infant could present with BESS subdural collections with very little trauma. Dr. Mack explained that it is not understood why some infants get BESS and others do not, or why it is more common in males. She stated that BESS is sometimes associated with a large head and is more common in premature infants.

Dr. Mack also described a network of blood vessels and cells in certain areas of the brain, called the plexus, which can bleed without trauma. She testified that it is not known why the dural plexus can bleed without trauma, and that doctors do not even understand why those vessels are there.

Dr. Mack testified that confirmatory tests are critically important because doctors cannot perform experiments, and that there is no confirmatory test for SBS/AHT. She stated that the lack of a confirmatory test affects the accuracy of the scientific literature, and that the literature on SBS/AHT is flawed because of a built-in confirmatory bias. She discounted data in the literature interpreted to show a 96 percent accuracy rate of retinal hemorrhages in diagnosed SBS/AHT cases.

Regarding D.J., Dr. Mack testified that there were no signs of trauma in D.J.'s scans: no soft tissue swelling, no skull fracture, and no abnormality within the brain itself. She described the importance of D.J.'s medical history as showing the subarachnoid space slowly expanding over time. She further

explained her opinion that D.J. had benign external hydrocephalus (BEH) -- subdural fluid collections in the context of enlarged subarachnoid spaces -- which are typically diagnosed only after children present with symptoms that often include seizures, lethargy or sleepiness, or not eating well.

In discussing D.J.'s retinal hemorrhages, Dr. Mack testified that retinal hemorrhages have been associated with BEH, sometimes being severe. Dr. Mack stated that if a pediatrician showed her scans like D.J.'s and claimed that retinal hemorrhages meant abuse, she would caution that the other relevant data -- no brain injury, enlarging subarachnoid space over time, and fluid collection -- cannot be ignored. She explained that assigning a label of abuse in a case like this one would be a circular argument that ignores the data on the imaging and sequence of films.

With regard to D.J.'s imaging, Dr. Mack explained that she would have called the emergency room to say that he had subdural hemorrhages and asked about any history to understand why. She further stated that it was appropriate that D.J.'s initial evaluation phase was reported as a potential abuse case. She agreed on cross-examination that shaking is dangerous and said that it could particularly cause injuries to the neck. Regarding D.J.'s February 13, 2017 image, Dr. Mack explained that she could not tell whether what occurred there was accidental or inflicted. She testified that there has not been good evidence

that shaking can cause bridging vein rupture, although there has been literature that presumes that it does.

On cross-examination, Dr. Mack agreed that she had never conducted a child abuse evaluation and that radiologists do not treat patients. In discussing BESS, the State contended that the scans prior to February 13, 2017, were normal. Dr. Mack disagreed, stating that the subarachnoid space slowly enlarged over time, so she would not call the scans normal, but they could reasonably be characterized as what can be seen among infants, and that the scans reflected BESS, a finding that was left untreated.

Dr. Mack testified that the premise that shaking alone could cause the injuries described in Dr. Medina's report was a heated controversy and that she did not think a finding of SBS/AHT could be made based on D.J.'s images.

d. Dr. Chris Alan Van Ee

Dr. Chris Alan Van Ee, an engineer who focuses on impact biomechanics and mechanical engineering, defined biomechanics as the application of mechanical principles to biological structures. He explained that impact biomechanics looks at the human body from a mechanical perspective to try to understand what forces or accelerations give rise to injury.

He testified that the forensic aspect involves looking at whether an injury could be identified as trauma and what acts could have produced it.

In discussing angular acceleration, Dr. Van Ee stated that angular acceleration -- which is how quickly the spin of something changes, here, the spin of the head -- could give rise to things like subdural hematoma or intracranial hemorrhage, i.e., hemorrhage inside of the skull around the brain. Dr. Van Ee explained that, for example, if a merry-go-round stopped abruptly, how quickly the stop occurred would be angular acceleration. He testified that the faster the stop occurs, the greater the angular acceleration, and the slower the stop occurs, the slower the angular acceleration.

Dr. Van Ee testified about how SBS/AHT relates to biomechanics. He stated that SBS/AHT is a hypothesis whereby if someone holds a child by the torso and shakes the child, the child's head would go back and forth, creating "angular accelerations of the head that are sufficient to rip bridging veins and cause injury to a child." Dr. Van Ee explained that if a biomechanics analysis were conducted, it would determine what those angular accelerations are and whether those angular accelerations are consistent or inconsistent with what we know causes injury in a car crash or in a fall. When asked whether biomechanics provides a way to test the hypothesis of shaking, he answered that science can be used to test it, but there is no perfect test.

Dr. Van Ee explained the distinction between a whiplash event and a shaking event, stating that Dr. Ommaya's 1968 study used a sled that "accelerate[d] th[e] chair with the primate in it at speeds representative of a 30-mile-per-hour crash" -- which was a whiplash event. He explained that in some whiplash events, for example if someone sitting at a red light was rear-ended by a vehicle traveling at five miles per hour, he could look at the head acceleration and compare it to a shaking event. He continued that in a low-speed incident like that, the accelerations could be five to 10 g, which does not compare to a 30-mile-per-hour crash. Dr. Van Ee noted that subdural hematomas in a five to 10 g whiplash event are typically not present, although he noted that there is always an exception to the rule, specifically if someone is particularly vulnerable. He provided the example of riding a roller coaster and explained that some people have developed subdural hematomas due to roller coasters, but that it was very rare that someone would suffer a traumatic injury from those types of exposures.

Dr. Van Ee testified that the hypothesis that shaking alone can result in damage to the bridging vein without simultaneously injuring the neck or torso cannot be scientifically supported. He stated that because the neck is very weak and vulnerable to injury, the first place to look for injury after a shaking motion, from a biomechanics standpoint, is the neck. He further noted that the

idea that a subdural hemorrhage could be caused by ripping a bridging vein without injuring the neck is unsupported by the data because the angular accelerations that are created in shaking are less than what is seen in even a one-foot fall.

Dr. Van Ee testified that if a baby fell over and hit their head on the carpet, such a fall would not be associated with a subdural hemorrhage or a massive traumatic brain injury, and that the angular accelerations of such a fall that have been measured and published are greater than what is seen in shaking. He explained that the data that exists in biomechanics shows that if someone shakes a child, the neck should be where injuries would start, and the levels of angular acceleration are well below the levels associated with head injury. Thus, he testified that it is not known whether the triad of symptoms and injuries can be produced by shaking alone, but neck injuries can be.

Addressing the Prange study published in the Journal of Neurosurgery, Dr. Van Ee stated that the authors developed a crash-test device that represented the weight and relative size of a six-week-old infant. He stated that researchers shook the test device, slammed it onto a piece of foam that represented a crib mattress, dropped it on the floor or the crib mattress from varying heights (one foot, three feet, and five feet), and slammed it against a wooden bench and also against a wooden bench with carpet on it. Dr. Van Ee

testified that the authors were trying to understand the head acceleration in each setting.

Dr. Van Ee explained the findings of the study, noting that the one-foot fall onto the floor gave rise to greater angular acceleration than acceleration resulting from shaking. He explained that the authors of the Prange study wrote that there was no data to indicate that shaking alone can give rise to the injuries associated with SBS/AHT. Defense counsel asked how those findings would change if an infant was larger, and Dr. Van Ee replied that the heavier the child is, the harder the child is to shake, so less overall head acceleration, angular velocity, and angular acceleration would be created.

Dr. Van Ee testified that the evidence is weak that shaking alone can actually cause the injuries associated with SBS/AHT. However, he explained that studies have shown that impact can cause those injuries. He stated that there are studies where children have died as a result of abuse and the autopsies reveal signs of impact that include subtle skull fracture or bleeding, bleeding of the scalp, swelling of the scalp, or other injuries.

Dr. Van Ee stated that he has not seen a biomechanical study that definitively concluded that shaking can cause the injuries associated with SBS/AHT. He stated that there has been no study conducted with dummies or animals that proved that subdural hematomas and retinal hemorrhages can be

caused by shaking alone. However, he acknowledged that there are confession articles in which people have confessed to shaking a child and the child had the injuries associated with SBS/AHT.

On cross-examination, Dr. Van Ee conceded that he is not a medical doctor, does not have any medical degrees, has not diagnosed or treated patients, has not been taught what a forensic examination for child abuse entails, has never been trained in diagnosing SBS/AHT, has never conducted an examination for abuse or been consulted when an examination for abuse occurred, and has never examined an infant before.

Dr. Van Ee explained that as of now “there is not a mechanistic explanation that allows one to go from shaking to those injuries” that makes sense. In discussing Dr. Duhaime’s study, Dr. Van Ee noted that approximately 15 children had died, and some had been thought to have been killed by shaking alone, but examiners who performed autopsies found evidence of impact in each child who died, so it was clear that the deaths did not result from shaking alone. Moreover, he testified that in the Duhaime study, football players shook a device that represented a one-month-old, and the accelerations measured were very low and below the level of what researchers thought would cause injury. Dr. Van Ee further explained that if someone violently shakes an infant, that action is going to cause serious injury

and maybe even death, but the question at issue is would the shaking result in the triad of symptoms at issue here and nothing else.

3. Trial Court Opinion

Following the hearing, on January 7, 2022, the trial judge issued a 75-page written decision and accompanying order granting Nieves's motion and barring Dr. Medina's expert testimony regarding SBS/AHT at trial.

The trial court concluded that SBS/AHT evidence was not reliable. The court explained that the diagnosis of SBS/AHT and the triad of symptoms associated with it are subject matters outside of the knowledge of the average juror, which "necessarily requires expert testimony to explain how an AHT diagnosis could be concluded without actual evidence of child abuse." The trial court ruled that it was clear from the literature and testimony that SBS/AHT "has never been medically [or] scientifically validated as a diagnosis because it has never been developed through scientific/medical techniques or procedures which, in turn, would make it a diagnosis that is scientifically or medically reliable."

The trial court further explained that the evidence presented, especially through the testimony of Dr. Medina, showed that SBS/AHT "is more conjecture than a diagnosis because it is an option embraced once a diagnostician runs out of diagnostic options." The court underscored that there

was no test cited by Dr. Medina or referenced in the literature that could support a finding that humans can produce the physical force necessary to cause the symptoms associated with SBS/AHT in an infant. It further noted that the State did not provide any evidence that Nieves inflicted any trauma upon D.J. and determined that SBS/AHT was a flawed diagnosis because it originates from a theory based upon speculation and extrapolation instead of being anchored in facts developed through reliable testing.

The trial court stated that the literature and testimony showed that SBS/AHT is an “assumption packaged as a medical diagnosis, unsupported by any medical or scientific testing, based upon scaled down versions of testing done on monkeys, wooden dolls, or other anthropomorphic surrogates.” The court noted that this assumption has been proffered in cases like this one as proof beyond a reasonable doubt as to the cause of an infant’s injuries. However, it found that there was no support to permit that type of proffer to stand because there has been no study that has ever validated the hypothesis that shaking a child can cause the triad of symptoms associated with SBS/AHT. The trial court stated that when used in a criminal courtroom to prove causation, SBS/AHT “can be highly prejudicial and far less probative . . . when no one has ever tested the capacity of an individual to shake a baby in an effort to cause the triad of symptoms defining” SBS/AHT.

The trial court observed that human babies are very different from the monkeys, wooden dolls, or other anthropomorphic surrogates utilized in the studies referenced and reviewed concerning the effects of force and impact. The court held that because of that, and as noted in Dr. Medina's testimony, we do not know, and will likely never know, what minimum force is necessary to cause subdural hematomas or the other triad of symptoms because the studies all reached diverse conclusions and, moreover, relied on models that cannot be deemed reliable mirrors of the infant brain.

The court explained that defendant's right to a fair trial includes the right to have only evidence that is sufficiently established as reliable under State v. Harvey, 151 N.J. 117 (1997), and Frye presented to a jury. The court noted that, on the strength of its name alone, SBS/AHT "evokes a sense of horror that affects the sensibilities of any competent juror, compromising their ability to follow the instructions of the court concerning the weighing of evidence fairly and impartially," but that there is no scientific technique or procedure to confirm SBS/AHT as a reliable diagnosis.

The trial court concluded that accordingly, SBS/AHT "cannot become part of a case unless coupled with physical evidence that an accused subjected an infant-victim to some impact of physical trauma that would support holding the accused criminally liable." Without such accompanying physical evidence,

the court determined, SBS/AHT “remains exactly what it is . . . a final option lacking a reliable diagnostic criteria masking as a diagnosis” and “a hypothesis based upon extrapolation of data, coupled with a ‘process of elimination’ engaged in by diagnosticians in an effort to reach a ‘conclusive diagnosis’ which, in the end, cannot be treated medically.” The court explained the danger of experts influencing jurors who hear the phrase “medical certainty” coming from an expert as suggesting a high degree of value and reliability concerning SBS/AHT as a proven diagnosis for causation.

Therefore, the court concluded that testimony concerning SBS/AHT could not be permitted in this case because it is not reliable evidence and was far more prejudicial than it was probative. The court explained that permitting such testimony in this case would be “the perfect recipe for a conviction not borne of a fair and unbiased decision-making process but, instead, one which would compromise the integrity of this prosecution and our criminal justice system.” The court determined that the State failed to provide evidence that Nieves inflicted force upon D.J. resulting in injuries symptomatic of an SBS/AHT diagnosis and failed to prove that the science behind SBS/AHT is sufficiently reliable to be used to implicate Nieves in abusive conduct and hold him criminally liable for causing D.J.’s injuries. Thus, the court granted Nieves’s motion to bar the admissibility of testimony concerning SBS/AHT.

Based on its decision barring SBS/AHT testimony, the trial court granted Nieves's motion to dismiss the indictment on January 28, 2022. The court held that "the State has insufficient evidence to prove causation in this case given the suppression of the testimony concerning [SBS/AHT]."

B. State v. Cifelli

Respondent Michael Cifelli is the father of J.C., who was born prematurely in November 2016. In late December 2016, when J.C. was 10 weeks old, Cifelli was caring for J.C. when he began to vomit excessively and exhibit fatigue and fever symptoms. Cifelli brought J.C. to the pediatrician and then ultimately to the hospital for treatment. J.C. was diagnosed with a viral illness and discharged the same day. The parents were told to follow up with the pediatrician and that J.C. might have a gastrointestinal illness.¹⁷

In early January 2017, while Cifelli was again caring for J.C., J.C. experienced seizure-like symptoms. J.C. was subsequently admitted to the hospital. While in the hospital, J.C. appeared again to be suffering from seizure-like symptoms, unaware of his surroundings, and not easily consolable. At the hospital, doctors determined that J.C. had fluid and blood around his

¹⁷ J.C.'s medical records are not in the record on appeal. The facts in this section are derived largely from Dr. Medina's testimony at a hearing regarding the DCPD proceedings in this matter, as well as the grand jury testimony.

brain and retinal hemorrhages in his left eye. Those findings triggered a child abuse investigation against Cifelli.

During J.C.'s hospitalization, his symptoms included the following: fluid around the brain that required surgery to drain; old and new brain bleeds; intraretinal and submacular retinal hemorrhages, meaning blood in multiple layers of his eyes; an apparent macular hole in his right eye, which was later diagnosed as "foveal vitreoretinal traction" -- marked by the gel-like substance, between the lens of the eye and the retina, pulling away from the retina; and a sudden increase in head circumference. J.C. underwent testing at the hospital. Dr. Medina reviewed his medical records and test results. Dr. Medina diagnosed J.C. with SBS/AHT with or without impact because she found "that there was no other medical diagnosis that could explain his symptoms."

On November 1, 2017, a Middlesex County grand jury indicted Cifelli on charges of aggravated assault and endangering the welfare of a child. Shortly thereafter, DCPD filed a Title 9 complaint against Cifelli, alleging that he caused J.C. to be an abused or neglected child. DCPD also filed an order to show cause seeking care and supervision of J.C. The complaint was dismissed

in August 2017, and Cifelli moved to dismiss the indictment.¹⁸ The trial court denied the motion.

On December 31, 2018, Cifelli moved for a Frye hearing to determine the admissibility of Dr. Medina's testimony regarding SBS/AHT, arguing that a diagnosis of SBS/AHT is scientifically unreliable. Judge Bucca denied Cifelli's motion, finding that SBS/AHT was generally accepted by the scientific community and was therefore reliable.

After the Appellate Division granted Nieves's motion for leave to appeal and remanded the matter for a Frye hearing, Cifelli moved for reconsideration of the denial in his case. Judge Bucca did not issue a written order, but the parties agreed to hold the matter in abeyance pending the outcome of the Frye hearing in the Nieves matter.

After the Frye hearing and Judge Jimenez's ruling barring expert SBS/AHT testimony in Nieves, Cifelli moved to dismiss his indictment, claiming the parties agreed to be bound by the Frye ruling in Nieves. The State opposed Cifelli's motion, arguing that while it agreed to await the outcome of the Frye ruling, it never agreed to be bound by the ruling.

¹⁸ Subsequently, on October 20, 2021, a second indictment was returned against Cifelli and J.C.'s mother, charging them with additional acts of child endangerment against both J.C. and his sibling. That indictment is not the subject of this appeal.

Judge Bucca determined that both sides had agreed to be bound by the ruling in Nieves. Pursuant to that agreement, Judge Bucca adopted Judge Jimenez's Frye ruling in Nieves and barred Dr. Medina's testimony regarding SBS/AHT in Cifelli's case. Judge Bucca, however, did not dismiss the indictment against Cifelli.

C. Consolidated Appellate Division Decision

The State moved for leave to appeal in both Nieves and Cifelli, arguing that it had established SBS/AHT's acceptance in the medical community and that trial judges therefore should not have barred its proposed expert testimony. State v. Nieves, 476 N.J. Super. 609, 617 (App. Div. 2023). The State also appealed the trial court's order barring testimony in Cifelli, arguing that it did not agree to be bound by the Nieves decision and that the Nieves decision was incorrect. Ibid. The Appellate Division granted leave to appeal in both matters, consolidated the appeals, and affirmed both trial court decisions. Id. at 617-18.

The Appellate Division held that "[t]he evidence supports the finding that there is a real dispute in the larger medical and scientific community about the validity of shaking only SBS/AHT theory, despite its seeming acceptance in the pediatric medical community." Ibid. The appellate court determined that the experts who testified at the Frye hearing all agreed that "there was

controversy surrounding the theory that the biomechanical principles underlying SBS/AHT actually supported the conclusion that shaking only can cause the injuries associated with SBS/AHT.” Id. at 618. Further, the court held that “[w]here the underlying theory integrates multiple scientific disciplines, as here, the proponent must establish cross-disciplinary validation to establish reliability,” which the State failed to do. Ibid.

The Appellate Division noted that there is a dearth of recent New Jersey cases challenging the admissibility of SBS/AHT testimony. Id. at 649. The court addressed several out-of-state SBS/AHT cases cited by the State, id. at 649-50, as well as the cases Nieves advanced, id. at 650-51. It noted that none of the cases cited by Nieves and only a limited number of the State’s cases expressly determined the admissibility of SBS/AHT evidence under Frye, and that others determined its admissibility under a different standard, or did not actually decide the question of the admissibility of SBS/AHT as a theory. Id. at 649-51. The court explained that although some of the cases cited by the State confirmed the reliability of SBS/AHT based on a prior court’s acceptance of SBS/AHT, a reliance on such cases would be “a type of circularity that is inappropriate given Nieves’s position that the medical and scientific community’s view about SBS/AHT has evolved over time, warranting a new review of the issue.” Id. at 651.

The court determined that SBS/AHT is a multidisciplinary diagnosis, and the question of whether it is accepted within the medical and scientific community “requires evaluation of two considerations: (1) whether the theory is generally accepted by the biomechanical community and supported by biomechanical testing; and (2) whether the theory is generally accepted by the pediatric medical community and supported by the clinical data connecting the constellation of symptoms with SBS/AHT.” Id. at 652.

The court held that the State “demonstrated general acceptance in the pediatric community,” but it agreed with the trial court that “the State [did] not demonstrate[] general acceptance of the SBS/AHT hypothesis to justify its admission in a criminal trial.” Ibid. To the contrary, the court found the evidence showed that there was “no general acceptance from the biomechanical community, and [that] biomechanical testing has never proven the premise of SBS/AHT, despite the hypothesis being grounded in biomechanical principles.” Id. at 652-53. Accordingly, the court affirmed the trial court’s decision precluding SBS/AHT testimony at trial. Id. at 654.

The court next addressed the State’s challenge to the trial court’s dismissal of Nieves’s indictment. Ibid. The court determined that “without SBS/AHT testimony, there was insufficient evidence to support the indictment against Nieves” because the State would not be able to prove the element of

causation under either the aggravated assault or endangering the welfare of a child charges. Id. at 655. The court explained that “[a]lthough the State could present testimony that D.J. was in Nieves’s care when D.J. had his episodes of limpness” and “was found to have retinal hemorrhages and subdural hematomas, the State would not be able to explain how Nieves harmed D.J., leaving the question for the jury to determine.” Ibid. The court noted that “a jury may draw a reasonable inference from the facts presented” but reasoned that permitting the State to rely on jury inference here would impermissibly “shift or lighten the burden of proof, or become a bootstrap to reduce the State’s burden of establishing the essential elements of the offense charged beyond a reasonable doubt.” Ibid. (quoting State v. Brown, 80 N.J. 587, 592 (1979)). The court explained that speculation cannot be disguised as a rational inference and that “[r]equiring the jury to infer that Nieves harmed D.J. would require the jury to make such a leap.” Ibid.

As to the Cifelli appeal, the court explained that because its decision resolved the admissibility of SBS/AHT testimony under Frye, and must be given conclusive weight, the decision of the trial judge in Cifelli to defer to the trial court’s Frye ruling in Nieves was moot. Id. at 657.

D. Review and Amicus Participation Granted

This Court granted the State’s petition for certification in Nieves and motion for leave to appeal in Cifelli. 256 N.J. 451 (2024).

Additionally, we granted the applications of the following individuals and entities to participate as amici curiae: the Attorney General of New Jersey and DCPD (jointly); the Public Defender of New Jersey, Office of Parental Representation; the Innocence Network and Center for Integrity in Forensic Sciences (jointly); a group of Medical Doctors writing in support of respondents;¹⁹ a group of biomechanical engineers;²⁰ a group of scholars whose work addresses cognitive bias;²¹ the Family Justice Resource Center, upEND Movement, MJCF Coalition, and Mothers Outreach Network (collectively); the American Academy of Pediatrics, collectively with seven

¹⁹ Dr. Jacob Andersson, Prof. Anders Eriksson, Dr. John Galaznik, Dr. Patrick Hamel, Dr. Ulf Högberg, Dr. Lawrence Hutchins, Dr. Charles J. Hyman, Prof. Niels Lynøe, Dr. Marvin Miller, Dr. David Ramsay, Dr. Cyrille Rossant, Dr. Robert K. Rothfeder, Dr. Irene Scheimberg, Dr. Joseph Scheller, Dr. Guillaume Sébire, Dr. Waney Squier, Dr. Dale Vaslow, Dr. Knut Wester, and Dr. R.K. Wright.

²⁰ Lindsay “Dutch” Johnson, Ph.D.; Ken Monson, Ph.D.; Kirk Thibault, Ph.D., D-IBFES; Keith Button, Ph.D., PE; and Johan Ivarsson, Ph.D.

²¹ Dr. Jeff Kukucka, Ph.D.; Keith A. Findley, Esq.; Dr. Deborah Davis, Ph.D.; and Dan Simon, Esq.

additional medical societies;²² a group of law and psychology professors who have studied false confessions;²³ and the Association of Criminal Defense Lawyers of New Jersey (ACDL).

IV. PARTIES' ARGUMENTS

A. State of New Jersey and Amici Supporting the State's Position

The State argues that Dr. Medina's diagnosis of D.J. was sufficiently reliable to be the subject of expert testimony under N.J.R.E. 702 and should have been admitted.

The State submits that it established SBS/AHT's general acceptance in the medical community, which is the relevant scientific community under Frye. Accordingly, it contends that controversy in the field of biomechanics does not justify the exclusion of a medical expert's testimony on SBS/AHT. The State submits that the Appellate Division erred in requiring general acceptance within both the pediatric medical community and the

²² New Jersey State Chapter of the American Academy of Pediatrics; American Association for Pediatric Ophthalmology and Strabismus; American Society of Pediatric Neurosurgeons; American Society of Pediatric Neuroradiology; American Professional Society on the Abuse of Children; Society for Pediatric Radiology; and The Ray E. Helfer Society.

²³ Dr. Richard A. Leo, Ph.D., J.D.; Dr. Hayley Cleary, M.P.P., Ph.D.; Dr. Kyle Scherr, Ph.D.; Dr. Lucy Guarnera, Ph.D.; Dr. Saul Kassin, Ph.D.; Dr. Lindsay Malloy, Ph.D.; Dr. Christian Meissner, Ph.D.; Dr. Allison Redlich, Ph.D.; and Dr. Melissa Russano, Ph.D.

biomechanical community because biomechanics is not of equal importance to pediatric medicine regarding the diagnosis of SBS/AHT and because Dr. Medina's diagnosis took relevant biomechanical research into account. The State submits that "the lack of general consensus on [SBS/]AHT in the biomechanical community is a result of biomechanical engineers' inability to replicate in a laboratory a phenomenon that clinicians have repeatedly observed in the real world." The State asserts that "[t]he fact that there exists some disagreement among biomechanical engineers generally regarding shaking alone as a mechanism of [SBS/AHT] should not render a diagnosis of [SBS/AHT] unreliable in a particular case" because "absolute scientific certainty is not the standard for the admissibility of expert testimony." The State further submits that biomechanical studies have failed to disprove that shaking alone can cause the intracranial injuries associated with SBS/AHT.

Moreover, the State argues that it established SBS/AHT's general acceptance in the medical community through the expert testimony of Dr. Medina, by offering authoritative scientific writings that establish the general acceptance of the SBS/AHT diagnosis and its underlying methodology in the medical community, and by offering numerous judicial opinions that have accepted SBS/AHT as a reliable scientific premise.

Additionally, the State asserts that Dr. Medina did not opine that shaking alone was the mechanism that caused D.J.’s injuries, but rather that D.J. suffered from SBS/AHT “that occurs with a shaking event with or without impact.” In the alternative, the State argues that the Court should remand these matters to a Special Adjudicator to conduct a comprehensive evidentiary hearing on the reliability of the SBS/AHT diagnosis under the standard announced in Olenowski I.

Several amici support the State’s position. The New Jersey Attorney General and DCPD argue that SBS/AHT is a generally accepted medical diagnosis long recognized by a host of national and international medical organizations. They maintain that the Appellate Division’s decision means the State may lose the ability to prosecute certain cases of child endangerment and the decision would also hamper DCPD’s ability to fulfill its Title 9 responsibilities. Amici echo the State’s argument that SBS/AHT satisfies the general-acceptance standard based on the presented expert testimony, scientific writings, and judicial opinions. They contend that biomechanics does not cut in either direction or resolve the dispute in anyone’s favor because it has not disproven the theory of SBS/AHT. They further argue that the Appellate Division’s reliance on State v. Pickett, 466 N.J. Super. 270 (App. Div. 2021), to support a finding of the need to consider more than one

scientific community is misplaced because it did not establish a bright-line rule.

The American Academy of Pediatrics and its joint participants (collectively, AAP) argue that the relevant scientific community for the diagnosis of abusive head trauma is the interdisciplinary specialties involved in the actual clinical evaluation, diagnosis, and treatment of patients. The AAP contends that SBS/AHT is a clinical medical diagnosis made by a clinician based on the signs, symptoms, and history of the patient. The AAP argues that because biomechanical engineers are not consulted as part of the diagnostic process, biomechanical engineering is not its own relevant scientific community regarding the clinical diagnosis of SBS/AHT. Moreover, the AAP argues that the medical community does not recognize a genuine debate of the validity of SBS/AHT.

B. Nieves and Cifelli and Amici Supporting Their Position

Respondent Nieves urges this Court to affirm the exclusion of expert testimony about SBS/AHT without impact injuries because the State failed to prove that the diagnosis is accepted within the relevant scientific communities of biomechanics and medicine. He argues that the State was required to prove the reliability of SBS/AHT (or SBS/AHT with no impact, and as distinguishable from AHT generally), which it failed to do. He contends that

SBS/AHT covers a wide range of injuries and symptoms and does not employ a specific diagnostic criterion; instead, doctors engage in a process of elimination to determine whether there are other explanations for the symptoms. Nieves submits that because SBS/AHT is a differential diagnosis that integrates multiple scientific disciplines, a finding of reliability must consider all fields used to render such a diagnosis. Nieves contends that because it was the biomechanical understanding of angular acceleration that led to the development of the SBS/AHT theory and the theory remains rooted in biomechanical principles, the relevant scientific community for purposes of a reliability analysis must include biomechanics.

Nieves argues that the State failed to prove that SBS/AHT is generally accepted as reliable in the field of biomechanics. He asserts that the State conceded and the record shows that SBS/AHT has never been validated in the field of biomechanics because no study has shown that humans can shake a child with enough force to cause the triad of symptoms associated with the SBS/AHT diagnosis. Additionally, he submits that research has shown that violent shaking would damage a child's neck before causing the triad of symptoms, which undermines the reliability of the SBS/AHT diagnosis.

Moreover, he contends that the State failed to prove that SBS/AHT is generally accepted as reliable in the field of medicine because it did not prove

that there is a generally accepted and reliable medical basis to believe that shaking can cause the triad without causing any other injuries. Instead, he asserts, the “general acceptance” the State proffered in the medical community relies on unsupported theories and unreliable confession studies. He also contends that the State failed to prove that SBS/AHT is generally accepted as reliable based on judicial authority.

Finally, Nieves argues that remand is unnecessary because appointing a Special Adjudicator will not affect whether the reliability of SBS/AHT is litigated in other cases, and an adequate record already exists because there was a Frye hearing below.

Respondent Cifelli generally relies on the arguments made by Nieves and contends that the judgments below should be affirmed. Cifelli also argues that the State has not established any error in the trial court’s ruling and should be barred by principles of judicial and equitable estoppel from challenging its own prior position on appeal. He maintains that the trial court found that the State agreed to be bound by the Nieves court’s Frye decision and that, to protect the integrity of the judicial process, the State should be barred under principles of estoppel from advocating a contrary position.

Several amici argue in support of respondents. The Office of the Public Defender argues that SBS/AHT is a medicolegal diagnosis for a child’s

pathology rooted in the kind of speculation and conjecture that this Court has long held to have no place in New Jersey's child welfare jurisprudence as the cause of harm or injury to a child.

The ACDL argues that admitting SBS/AHT diagnoses in criminal court would lower New Jersey's high standards for the reliability of evidence. The ACDL contends that the biomechanical scientific community's rejection of SBS/AHT as a diagnosis undermines the reliability of the testimony at issue here.

The biomechanical engineers argue that experts in biomechanics are members of the relevant scientific community for determining the reliability of SBS/AHT because the diagnosis is rooted in biomechanical principles regarding the effects of rotational forces on the contents of the skull. They explain that biomechanical expertise is necessary to answer the question of whether it is even physically possible for a human adult to shake an infant in a manner that can produce forces sufficient to cause the triad of symptoms. The biomechanical engineers contend that the current state of biomechanical research does not support the validity of an SBS/AHT-by-shaking diagnosis because such research does not support the theory that shaking can cause the triad of symptoms associated with an SBS/AHT diagnosis. The biomechanical engineers maintain that, in light of those biomechanical findings and other

research, courts and medical professionals are increasingly skeptical of SBS/AHT diagnoses.

The medical doctors supporting respondents argue that there is no scientific basis for an SBS/AHT diagnosis. They contend that there are no scientific or biomechanical studies that validate the hypothesis that abusive shaking can cause the triad of symptoms; that every component of the triad has numerous non-abusive causes; and that SBS/AHT is not a medical diagnosis. They contend that the SBS/AHT hypothesis was adopted based on the misapplication of animal studies and circular reasoning. They submit that the shift to evidence-based medicine exposed the dearth of reliable literature supporting the SBS/AHT hypothesis. The medical doctors argue that the SBS/AHT hypothesis is not generally accepted in the relevant scientific community, which includes biomechanical scientists.

The cognitive bias scholars argue that there are high error rates in subjective medical determinations like SBS/AHT which are exacerbated by cognitive bias. They maintain that context bias especially impacts SBS/AHT determinations, as experimental studies clearly establish.

Family Justice Resource Center, upEND Movement, MJCF Coalition, and Mothers Outreach Network argue that children and parents experience serious long-term harm to their mental and physical health as a result of forced

separation -- which repeatedly occurs on the basis of unreliable scientific evidence.

The Innocence Network and Center for Integrity in Forensic Sciences (jointly, Innocence Network) argue that the trial and appellate courts properly excluded the testimony regarding SBS/AHT by following the current scientific understanding rather than outdated legal precedent. The Innocence Network maintains that the current scientific consensus does not support diagnoses of SBS/AHT based only on the triad of symptoms associated with shaking. The Innocence Network argues that courts across the country are thus increasingly recognizing that the SBS/AHT diagnosis is not supported by science. It contends that recognition of the new scientific understanding of SBS/AHT has led to numerous reversals and exonerations. The Innocence Network maintains that forensic evidence is uniquely powerful in its ability to influence and potentially mislead jurors and factfinders, and that the risk of admitting flawed forensic evidence is not hypothetical, but the leading cause of wrongful conviction.

The professors who study false confessions argue that the SBS/AHT “confession studies” the State relies on fail to account for the well-documented phenomenon of false confessions and improperly assume that the purported confessions were true without independent corroboration.

V. LEGAL STANDARDS

A. Standard of Review

“Generally, appellate courts review a trial court’s determination of the admissibility of evidence for an abuse of discretion.” Harvey, 151 N.J. at 166. However, “[w]hen reviewing a decision on the admission of scientific evidence, an appellate court should scrutinize the record and independently review the relevant authorities, including judicial opinions and scientific literature.” Id. at 167. Additionally, “[w]hether expert testimony is sufficiently reliable to be admissible under N.J.R.E. 702 is a legal question we review de novo.” State v. J.L.G., 234 N.J. 265, 301 (2018).

B. Expert Testimony

N.J.R.E. 702 governs the admission of expert testimony. The rule provides that “[i]f scientific . . . knowledge will assist the trier of fact to understand the evidence or to determine a fact in issue, a witness qualified as an expert by knowledge, skill, experience, training, or education may testify thereto in the form of an opinion or otherwise.” N.J.R.E. 702. The proponent of expert testimony must establish that “(1) the subject matter of the testimony [is] ‘beyond the ken of the average juror’; (2) the field of inquiry ‘[is] at a state of the art such that an expert’s testimony could be sufficiently reliable’; and

(3) ‘the witness [has] sufficient expertise to offer the’ testimony.’ J.L.G., 234 N.J. at 280 (quoting State v. Kelly, 97 N.J. 178, 208 (1984)).

During the relevant period, New Jersey courts relied on the Frye standard to assess the reliability prong. Ibid. The Frye standard requires trial courts “to determine whether the science underlying the proposed expert testimony has ‘gained general acceptance in the particular field in which it belongs.’” Ibid. (quoting Frye, 293 F. at 1014). Notwithstanding the Frye test’s emphasis on general acceptance in the relevant field(s), the science underlying the expertise “must have a ‘sufficient scientific basis to produce uniform and reasonably reliable results [that] will contribute materially to the ascertainment of the truth.’” State v. Chun, 194 N.J. 54, 91 (2008) (quoting State v. Hurd, 86 N.J. 525, 536 (1981)).

Reliability of the expert testimony is of paramount import, and “[i]n evaluating the admissibility of scientific evidence introduced by the State in criminal cases, it is important to recognize that a high degree of reliability is necessary where the freedom, or even the life, of an individual is at stake.” Biunno, Weissbard & Zegas, Current N.J. Rules of Evidence, cmt. 6.1 on N.J.R.E. 702 (2025). “‘Proof of general acceptance within a scientific community can be elusive,’ and ‘[s]atisfying the test involves more than simply counting how many scientists accept the reliability of the proffered

[technique].” State v. Cassidy, 235 N.J. 482, 492 (2018) (alterations in original) (quoting Harvey, 151 N.J. at 171). Further, because labeling evidence as “scientific” or “expert” creates a “clear” “danger of prejudice” -- i.e., the possibility that juries will “accord excessive weight to unreliable expert testimony” simply because of its label, State v. Cavallo, 88 N.J. 508, 518 (1982) -- “[g]eneral acceptance ‘entails the strict application of the scientific method, which requires an extraordinarily high level of proof based on prolonged, controlled, consistent, and validated experience.’” Cassidy, 235 N.J. at 492 (quoting Harvey, 151 N.J. at 171).

The proponent of the expert evidence has the burden to “clearly establish” general acceptance, Cassidy, 235 N.J. at 492; Chun, 194 N.J. at 92, in one or more of three ways: “expert testimony, authoritative scientific and legal writings, and judicial opinions,” J.L.G., 234 N.J. at 281; see also Cassidy, 235 N.J. at 492. Courts, in turn, act as gatekeepers for ensuring the reliability of expert testimony. J.L.G., 234 N.J. at 307-08; see also State v. Sowell, 213 N.J. 89, 99-100 (2013).

A key step in a court’s gatekeeping function under Frye is to identify the relevant scientific community. See Pickett, 466 N.J. Super. at 302. Importantly, in certain circumstances, “there might be more than one scientific community to consider.” Ibid. And, when there is more than one relevant

scientific community to consider, the proponent of the expert evidence must clearly establish general acceptance in all of the relevant scientific communities involved. See ibid. Specifically, “cross-disciplinary validation” may be required “to determine reliability” when validation by a single scientific community is “too narrow” because there is another scientific community or communities “to which [the evidence also] belongs.” Id. at 323. Indeed, in this Court’s opinion in Olenowski II, we recognized that the relevant scientific communities for determining the reliability of Drug Recognition Expert (DRE) testimony included both medicine and toxicology. State v. Olenowski (Olenowski II), 255 N.J. 529, 604 (2023).

This Court has on many occasions determined the admissibility of expert testimony under Frye. Recently, in J.L.G., we revisited a previously accepted theory as the science developed and determined that it no longer met the reliability prong and was therefore inadmissible in criminal trials because it was based on clinical practice and not supported by objectively tested scientific evidence. 234 N.J. at 272, 291. There, the defendant challenged the scientific reliability of expert testimony regarding Child Sexual Abuse Accommodation Syndrome (CSAAS). Id. at 272. CSAAS is a diagnostic tool comprised of five categories of behavior commonly demonstrated by child sex

abuse victims: “secrecy; helplessness; entrapment and accommodation; delayed, conflicted, unconvincing disclosure; and retraction.” Id. at 271.

Although we had previously found CSAAS expert testimony reliable in 1993, and it had been presented in child sexual abuse trials throughout the country for over twenty years, we determined that “[b]ased on what is known today, it is no longer possible to conclude that CSAAS has a sufficiently reliable basis in science to be the subject of expert testimony.” Id. at 272. We found significant disagreement among the scientific community on the issue of false recantation because recantation rates varied among studies, and although researchers agreed that recantation happens in a minority of cases, they disagreed on the rate. Id. at 295-97. Additionally, we found denial was a heavily debated topic and that the studies that found higher rates of denial were of questionable validity. Id. at 297-99.

In applying the Frye standard, we held that “expert testimony about CSAAS in general, and its component behaviors other than delayed disclosure, may no longer be admitted at criminal trials.” Id. at 272. In so holding, we acknowledged that “CSAAS stems from observations made in clinical practice -- not systematic scientific study,” and that while “[c]linical wisdom is valuable, . . . it must be examined with care and objectively tested.” Id. at 291 (emphases added). We determined that there was not general acceptance in the

scientific community about either retraction or false denial. Id. at 302.

Additionally, we explained that “[t]o satisfy the reliability prong . . . it is not enough to state that certain behaviors can be observed in some victims some of the time -- or that those behaviors are not inconsistent with abuse” because “[t]hat is not the type of ‘state of the art’ evidence the case law requires.” Id. at 303 (quoting Kelly, 97 N.J. at 208).

C. Case Law Addressing SBS/AHT

As the Appellate Division noted, few New Jersey court opinions have addressed the reliability of SBS/AHT, and none involve the reliability of SBS/AHT in the context of shaking without impact. Two prior New Jersey cases ruled on the admissibility of SBS/AHT in cases involving shaking with impact: State v. Compton, 304 N.J. Super. 477 (App. Div. 1997), and State v. Galloway, 133 N.J. 631 (1993).

In Compton, a defendant’s four-and-one-half-month-old son died in his care, and the defendant was charged with murder. 304 N.J. Super. at 479, 481. While in custody, the defendant told his wife that he had thrown the infant in the air, and the infant went through his hands and fell. Id. at 482. The defendant then shook the infant to try to revive him. Ibid. On appeal, the defendant challenged, among other things, the general acceptance and reliability of the SBS/AHT diagnosis. Id. at 483. The Appellate Division

analyzed the testimony of the State’s expert, the medical research and literature, and case law from other jurisdictions. Id. at 485-87. The court determined that SBS/AHT evidence was admissible because the State’s expert testified that the theory was generally accepted, the condition had been adequately analyzed and recognized in medical research and literature, and other jurisdictions had recognized the theory. Ibid. Moreover, the Appellate Division noted that this Court, in Galloway, “recognized the condition implicitly, by acknowledging expert testimony describing [SBS/AHT] in connection with a particular case at bar, or treating it as an accepted medical condition without further comment.” Id. at 486.

In Galloway, the defendant was convicted of the murder of his girlfriend’s three-month-old son based on SBS/AHT evidence. 133 N.J. at 637. The defendant stated that he had picked up the baby, who was crying, but fell while carrying him, causing the infant to cry more. Id. at 638. The defendant admitted that he then shook the baby hard several times to stop the crying. Ibid. The defendant challenged his conviction based on, among other things, the trial court’s jury instruction regarding diminished capacity. Id. at 637. In describing the facts of the case, the opinion noted that the defendant’s shaking of the infant “caused the child’s head to bob back and forth rapidly, causing hemorrhaging of the blood vessels of the child’s brain, commonly

known as the ‘shaken baby syndrome.’” Id. at 638. We reversed and remanded based on the jury instruction and its interpretation of the child endangerment statute. Id. at 649, 662. We did not analyze SBS/AHT or mention it again in the opinion. See generally id. (including no analysis or other mention of SBS/AHT).

Although New Jersey case law on SBS/AHT is sparse, many cases from around the country in the past 15 years have focused on the reliability of SBS/AHT. See Ex parte Roark, 707 S.W.3d 157, 185, 187-88 (Tex. Crim. App. 2024) (granting the defendant’s application for a writ of habeas corpus, vacating his conviction, and remanding for a new trial because scientific knowledge regarding the State’s SBS/AHT theory of injury had evolved since the time of trial, and the defendant likely would not have been convicted if such newly evolved scientific evidence had been presented at trial); Smith v. State, 882 S.E.2d 300, 302, 308-10, 312 (Ga. 2022) (vacating the trial court’s order denying the defendant’s extraordinary motion for new trial and remanding for an evidentiary hearing to determine whether the defendant’s proposed expert testimony regarding developments in the understanding of the triad of symptoms and the diagnosis of SBS/AHT constituted newly discovered evidence); Allison v. State, 448 P.3d 266, 267-68, 271-74 (Alaska Ct. App. 2019) (reversing the defendant’s conviction because the trial court

erred in barring the defendant's genetic medical testimony as a defense against SBS/AHT allegations); Commonwealth v. Millien, 50 N.E.3d 808, 809-10 (Mass. 2016) (holding that the defendant was denied the right to effective assistance of counsel because his attorney did not retain a medical expert when the prosecution's case rested almost entirely on medical expert testimony, and noting that "[t]here is a heated debate in the medical community as to whether a violent shaking of a baby alone can generate enough force to cause the triad of symptoms of traumatic brain injury, and as to whether these symptoms can sometimes be caused by a short accidental fall"); People v. Bailey, 41 N.Y.S.3d 625, 626-27 (App. Div. 2016) (affirming the trial court's decision to vacate the defendant's conviction and grant a new trial because advancements in science and medicine constituted newly discovered evidence calling into question the validity of the SBS/AHT diagnosis); Del Prete v. Thompson, 10 F. Supp. 3d 907, 954, 957, 958 n.10 (N.D. Ill. 2014) (finding that the defendant had carried her burden in petitioning for habeas relief, establishing that no reasonable jury would find her guilty beyond a reasonable doubt through new SBS/AHT testimony regarding the science of biomechanics and the fact "that the evidence basis for the proposition that shaking alone can cause injuries of the type at issue here is arguably non-scientific," and noting that "more recent developments in this area . . . arguably suggest[] that a claim

of shaken baby syndrome is more an article of faith than a proposition of science”); State v. Edmunds, 746 N.W.2d 590, 596-98 (Wis. Ct. App. 2008) (reversing the denial of the defendant’s motion for a new trial on the basis that “newly discovered evidence in this case shows that there has been a shift in mainstream medical opinion since the time of [the defendant’s] trial as to the causes of the types of trauma” that the deceased infant exhibited); Cavazos v. Smith, 565 U.S. 1, 9, 13 (2011) (Ginsburg, J., dissenting) (dissenting from the United States Supreme Court’s summary reversal and remand of the Ninth Circuit’s grant of a habeas petition, despite the Court finding that the jury’s verdict was supported by the record, in part because “[d]oubt has increased in the medical community ‘over whether infants can be fatally injured through shaking alone’” (quoting Edmunds, 746 N.W.2d at 596)).

The Michigan Supreme Court recently addressed the reliability of SBS/AHT testimony in People v. Ackley, 870 N.W.2d 858, 859-60 (Mich. 2015). In that case, the court determined whether a defendant was denied effective assistance of counsel when his trial counsel failed to secure an expert witness to defend against SBS/AHT accusations. Ibid. The defendant was convicted of first-degree murder and first-degree child abuse after a three-year-old in his care died. Id. at 860. At trial, defense counsel reached out to only one expert, a doctor, who “advised counsel . . . that he was ‘not the best

person' for the defense.” Ibid. The doctor stated that, for the purpose of assisting the defense, he was on the wrong side of the “religion”-like divide between doctors when it came to diagnosing SBS/AHT and injuries that result from short falls, meaning he could not assist the defense. Id. at 860-61.

However, counsel did not contact or retain another expert and continued to rely on that doctor in preparing for trial. Id. at 861. The court granted a new trial, and the appellate court reversed that determination on appeal. Id. at 862.

The Michigan Supreme Court reversed the appellate court’s ruling. Id. at 867. It determined that counsel’s efforts did not satisfy his duty to make reasonable investigations, “especially in light of the prominent controversy within the medical community regarding the reliability of SBS/AHT diagnoses.” Id. at 864. It explained that counsel’s failure to secure an expert to assist in the defense undermined the court’s confidence in the outcome of the case because “in a SBS/AHT case . . . where there is ‘no victim who can provide an account, no eyewitness, no corroborative physical evidence and no apparent motive to kill,’ the expert ‘is the case.’” Id. at 867 (quoting Deborah Tuerkheimer, The Next Innocence Project: Shaken Baby Syndrome and the Criminal Courts, 87 Wash. U. L. Rev. 1, 27 (2009)). Thus, the Michigan Supreme Court found that the failure of the defense counsel to even attempt to

find a supportive expert was ineffective assistance that likely impacted the case. Ibid.

More recently, the Michigan Supreme Court substantively explored the admissibility of expert testimony and the controversy surrounding the diagnosis of SBS/AHT in People v. Lemons, 22 N.W.3d. 42 (Mich. 2024). In that case, the defendant was convicted of first-degree felony murder in 2005 for the death of an infant following an allegation of SBS/AHT. Id. at 48. Subsequently, the defendant filed a motion for relief from judgment, “arguing that new evidence undermined the prosecution’s theory that the child’s cause of death was” SBS/AHT. Ibid. The trial court held evidentiary hearings and determined that the defendant’s proposed biomechanical engineering expert testimony was inadmissible under Michigan’s Daubert-like expert witness testimony rule. Ibid.

The Michigan Supreme Court disagreed and stated that “the SBS[/AHT] hypothesis is inherently ‘grounded in biomechanical principles.’” Id. at 55 (quoting Nieves, 476 N.J. Super. at 653). It held that the testimony by the defendant’s biomechanical expert (Dr. Van Ee, who participated in the Frye hearing in this matter) satisfied the rule and was thus admissible because he “was a qualified expert in the field of biomechanical engineering”; “[h]is testimony regarding the biomechanical mechanism of SBS[/AHT] would assist

the trier of fact in ascertaining a fact at issue”; the field of “[b]iomechanical engineering is a legitimate field of scientific study”; his “testimony was ‘based on sufficient facts or data’ and was ‘the product of reliable principles and methods’”; and he was able to apply those principles and methods reliably to the facts of the case. Id. at 56, 57 (quoting Mich. R. Evid. 702).

Importantly, the Michigan Supreme Court acknowledged that “a segment of the medical community takes significant issue with biomechanical research on SBS[/AHT],” but it determined that, “just as a biomechanical engineer may not testify about medical causation outside of their expertise, the medical community is not the judge of the validity of biomechanical research, nor is it the sole relevant expert community with respect to SBS[/AHT].” Id. at 59. It found “the position that biomechanics -- the study of forces acting on and generated within the human body -- is divorceable from a diagnosis of [SBS/AHT] to be untenable.” Ibid.

There is, of course, also case law finding SBS/AHT evidence reliable and therefore admissible at trial. See Clark v. State, 315 So. 3d 987, 993, 995, 996-97 (Miss. 2021) (holding that SBS/AHT expert testimony was reliable and properly admitted); Nielsen v. State, 430 P.3d 740, 751 (Wyo. 2018) (noting that the differential diagnosis used to diagnose SBS/AHT is a sufficiently reliable methodology, and that the SBS/AHT experts in the case “properly

explained their diagnoses . . . and how they arrived at their conclusions.”); Wolfe v. State, 509 S.W.3d 325, 327, 335, 341 (Tex. Crim. App. 2017) (affirming defendant’s conviction and holding that SBS/AHT testimony was properly admitted at trial because it was sufficiently reliable); People v. Flores-Estrada, 51 N.Y.S.3d 863, 864-65 (Sup. Ct. 2017) (denying defendant’s motions to preclude expert SBS/AHT testimony and for a Frye hearing regarding SBS/AHT); People v. Thomas, 998 N.Y.S.2d 590, 592 (Cnty. Ct. 2014) (same); Futrell v. Commonwealth, 471 S.W.3d 258, 265, 281-86 (Ky. 2015) (reversing defendants’ convictions because the trial court abused its discretion by refusing to remove, for cause, two unqualified jurors, but holding that the trial court did not abuse its discretion in admitting SBS/AHT expert testimony); Johnson v. State, 933 So. 2d 568, 570 (Fla. Dist. Ct. App. 2006) (affirming defendant’s conviction and holding that the medical expert’s SBS/AHT opinion was admissible, and that a Frye hearing was unnecessary).

VI. ANALYSIS

With those legal principles in mind, as well as the history of SBS/AHT and the numerous studies and articles presented by both parties, we turn to whether SBS/AHT expert testimony should be admitted in cases governed by

the Frye standard in which defendants are charged with child abuse offenses as a result of shaking without impact.

A. The Relevant Scientific Communities

We must first determine the relevant scientific community or communities for purposes of SBS/AHT.

As Dr. Medina testified, the starting point in the evolution of SBS/AHT was Dr. Ommaya's 1968 whiplash study, which tested the acceleration and deceleration of monkeys' brains during a simulated rear-end car accident at 30 miles per hour. Drs. Guthkelch and Caffey later relied on Dr. Ommaya's study in conceptualizing SBS/AHT. Dr. Caffey's conclusion that SBS/AHT, which he called infant whiplash syndrome, could result from shaking a child was based in part on Dr. Guthkelch's paper, which, in turn, was based on Dr. Ommaya's study.

It is therefore evident that the foundation of SBS/AHT lies in biomechanical science and engineering. The State and its supportive amici, including the AAP, note that the biomechanical engineering community informs the SBS/AHT inquiry but protest that biomechanical engineers are not of significant relevance or are not a scientific community unto themselves. Indeed, the State notes that "[SBS/]AHT is a clinical diagnosis that is informed by research in biomechanics" but at the same time argues that the

biomechanical community is “not of coequal importance” in the debate regarding the reliability of SBS/AHT expert testimony. What’s more, the State proffered no biomechanical expert testimony on its behalf.

It seems particularly odd, knowing that the origins of SBS/AHT lie entirely in biomechanics, to relegate biomechanics to second-class status. It cannot be that the scientific community upon which the very foundation of the State’s SBS/AHT testimony is based is not as relevant to the theory’s acceptance as the medical community that now diagnoses SBS/AHT -- a diagnosis that quite possibly would not have existed if not for a biomechanical study and the field of biomechanics in the first place.

Moreover, a scientific community is either relevant or not for purposes of determining admissibility of scientific evidence at trial -- degrees of relevance are not weighed, and for good reason. Were this Court to agree that the biomechanical engineering community is a relevant community -- but not that relevant -- the question would then arise regarding how much relevance our courts are to afford to this relevant community. Would testimony from an expert in an only “somewhat relevant” community be presented to a jury, with the jury instructed to afford only 20 or 30 percent weight to that testimony? That is not a workable procedure. As the Michigan Supreme Court noted in

Lemons, it is simply untenable to divorce the diagnosis of SBS/AHT from its origin, the study of biomechanics.

There can certainly be more than one relevant scientific community for purposes of Frye. Again, in Olenowski II, we recognized that the relevant scientific communities in determining the reliability of DRE testimony were both the medical and toxicology communities. 253 N.J. at 604. Here, we find that the relevant scientific communities for purposes of determining the reliability of SBS/AHT expert testimony are both the medical/pediatric community and the biomechanical engineering community.

B. Application of the Frye Standard of Reliability

We now turn to whether, under the Frye standard of reliability, the State has met its burden of establishing that Dr. Medina's expert testimony regarding SBS/AHT without impact is sufficiently reliable for admission at trial. The State, as the proponent of the expert evidence, has the burden to "clearly establish" general acceptance in the relevant scientific communities through expert testimony, authoritative scientific and legal writings, and judicial opinions. See J.L.G., 234 N.J. at 281; Cassidy, 235 N.J. at 492. Based on the testimony, evidence, and scientific studies and writings, this Court finds that the State has not met that burden. We find that the State has not shown general acceptance within the biomechanical community regarding whether

shaking without impact can produce the “triad” of symptoms associated with SBS/AHT: subdural hematomas, multilayered retinal hemorrhages, and encephalopathy.

We address the State’s argument that the Appellate Division misunderstood Dr. Medina’s diagnosis and improperly focused its opinion on shaking without impact. Specifically, the State argues that Dr. Medina diagnosed D.J. with SBS/AHT “as it occurs with a shaking event with or without impact.” Although, the State is correct that Dr. Medina noted that diagnosis in her records, she testified on cross-examination that there was no indication that D.J.’s head hit against anything and that D.J. did not “have any external signs of impact.” Therefore, it is evident from Dr. Medina’s own testimony that she did not observe any signs of any impact, so Nieves’s case is properly assessed as one involving shaking without signs of impact, as Dr. Medina herself testified. Although doctors may use the phrase “shaking with or without impact” as a catchall, those events are, as the studies and science make clear, very different and should be treated as such.

Regarding the scientific foundation of SBS/AHT, Dr. Ommaya’s 1968 whiplash study concluded, as explained above, that brain injuries could result from “rotational displacement of the head on the neck alone, without significant direct head impact.” Ommaya, 204 J. Am. Med. Assoc. at 285. As

Dr. Medina herself testified, Dr. Ommaya's study was later referenced as the scientific and biomechanical basis for what eventually became known as SBS/AHT. But Dr. Ommaya's study had nothing to do with infants or the understanding of what happens in the shaking of infants. His study focused on recreating whiplash events in monkeys at speeds of 30 miles per hour that mimicked rear-end car collisions. Drs. Guthkelch and Caffey used Dr. Ommaya's whiplash study to support their hypothesis that young children with brain injuries were subject to abusive shaking.

Over three decades after he published his whiplash study, Dr. Ommaya published a follow-up article questioning and challenging the scientific basis of SBS/AHT and specifically noting that Drs. Guthkelch and Caffey failed to factor into their theories the level of speed employed during his whiplash study. More pointedly, Dr. Ommaya stated that the assumptions upon which pediatric brain and retinal literature had been based in the past few decades -- assumptions based on his own study -- were "ambiguous or incorrect" and were used in the differential diagnosis of SBS/AHT "usually without reference to available biomechanical analysis." Ommaya et al., 16 Brit. J. Neurosurgery at 227.

Thus, Dr. Ommaya, whose 1968 study was the foundation of SBS/AHT theory and diagnosis in the first instance, publicly disclaimed the validity of

SBS/AHT's foundation. Further, he specifically noted that much of the literature and study of SBS/AHT was made without reference to available biomechanical analysis, a field of study that is imperative to understanding the manner in which acceleration and deceleration forces occur within the body. Indeed, even Dr. Caffey's 1974 study, which is universally referenced regarding the beginnings of SBS/AHT theory, concluded that the evidence at the time was "manifestly incomplete and largely circumstantial." Caffey, 54 Pediatrics at 403. And then Dr. Guthkelch, over 40 years after he published a study advancing the theory of SBS/AHT, also questioned the scientific foundation of that diagnosis. Guthkelch, 12 Hous. J. Health L. & Pol'y at 202-04. Dr. Guthkelch further called for "civility in scientific discourse" because he believed the controversy regarding SBS/AHT had risen to a level of divisiveness that interfered with the commitment to pursuing the truth. Id. at 201.

Indeed, this debate regarding the scientific validity of SBS/AHT without impact has been extraordinarily contentious. For example, the AAP and many other medical organizations around the world adopted the Consensus Statement, which employed the terms "denialists" and "denialism" to describe those who have questioned the scientific basis for SBS/AHT, and noted that "denialism of child abuse has become a significant medical, legal and public

health problem.” Choudhary et al., 48 Pediatric Radiology at 1050, 1059. The charged language used in the discourse surrounding SBS/AHT on both sides is disconcerting, given that the goal is, and should be, a search for truth and understanding.

The record established in the Frye hearing regarding the challenges to the science behind SBS/AHT in the matters before us does not deny the existence of child abuse, but rather questions the scientific and biomechanical underpinnings of the diagnosis of SBS/AHT as the sole basis for a legal finding of child abuse. Indeed, even the experts who testified for the defense at the Frye hearing acknowledged that shaking a young child could result in injury. Dr. Scheller specifically said that shaking a baby six months or younger would not be “a good thing” but questioned exactly what happens when a child is shaken and whether the science supports the triad of symptoms as resulting from shaking alone. Dr. Mack testified that she believed it was appropriate for D.J.’s initial evaluation at the hospital to be reported as a potential abuse case and agreed that shaking is dangerous, but she explained that child abuse cannot simply be assumed when the science supports many potential causes for the symptoms associated with SBS/AHT. Such questions regarding the scientific reliability of an SBS/AHT diagnosis, in search of truth

and understanding in the ever-evolving realm of science, are appropriate scientific inquiries.

As the trial court found, the evidence presented at the hearing, including Dr. Medina's testimony, showed that there was no test supporting a finding that humans can produce the physical force necessary to cause the symptoms associated with SBS/AHT in a child. The trial court further found SBS/AHT to be a theory not anchored in facts developed through reliable, scientific testing to confirm the diagnosis. This Court acknowledges that humans cannot be used as test subjects in potentially dangerous experiments regarding SBS/AHT, so there may never be a perfect scientific test. It is true that there are limitations to researching and understanding SBS/AHT and the forces necessary to result in the injuries associated with the diagnosis. But those limitations should not give way to assumptions in making an SBS/AHT diagnosis for which significant criminal liability follows when no other evidence of abuse is present.

The Consensus Statement noted that "[b]ecause medicine and science are dynamic, it is important to continually evaluate new hypotheses and, consequently, reevaluate previously confirmed scientific understanding, thus avoiding a rush to judgment." Choudhary et al., 48 Pediatric Radiology at 1055. Yet the Consensus Statement itself does not appear open to the

reevaluation of the very science upon which SBS/AHT rests. The Consensus Statement briefly mentioned Dr. Ommaya's study in noting the earlier studies that Dr. Caffey used to "show the effects of rotational acceleration/deceleration of whiplash." Id. at 1051. But the Consensus Statement says nothing further regarding Dr. Ommaya and the foundational elements of his work used by Dr. Caffey to craft SBS/AHT theory. Again, Dr. Ommaya's later study criticized the misuse of his own work as the scientific basis for SBS/AHT, noting that the assumptions upon which SBS/AHT retinal and brain injuries have been based "are ambiguous or incorrect" and made "usually without reference to available biomechanical analysis."

One cannot overlook that the biomechanical community is the scientific field from which SBS/AHT originated. A biomechanical study -- Dr. Ommaya's in 1968 -- was deemed by the medical community sufficient to form the basis for SBS/AHT theory, and biomechanics as a science in general is important in areas involving experiments of crash-test dummies to aid in making vehicles and vehicle parts safer. Dr. Medina even testified that "what we know about shaking, and the established thresholds for intracranial injury comes from that study" and that "everything else in biomechanics is based on th[e] thresholds" from Dr. Ommaya's study. Thus, SBS/AHT's underlying

basis in biomechanics plays a significant role in determining the reliability of SBS/AHT testimony.

Proof of general acceptance within a scientific community, here both the medical and biomechanical communities, “involves more than simply counting how many scientists accept the reliability of the proffered [technique],” Cassidy, 235 N.J. at 492, and certainly more than blindly accepting one community’s claim that there is no issue regarding general acceptance to begin with. The State noted in its supplemental brief that “there exists some disagreement among biomechanical engineers generally regarding shaking alone as a mechanism of AHT.” That concession alone, the lack of consensus in one of the relevant scientific communities, is itself strong evidence that there is no general acceptance in the biomechanical community of the SBS/AHT without impact diagnosis.

Contrary to the dissent’s assertion that the trial court, the Appellate Division, and this Court have merely relied on the individual views of six biomechanical engineers who either testified below or submitted briefing to this Court, as noted, the State itself has acknowledged the lack of scientific consensus among biomechanical engineers regarding shaking alone. Dr. Medina also testified that there is controversy in biomechanics focused on whether shaking can cause the forces needed to generate intracranial injury on

infants. The dissent does not discuss Dr. Ommaya's 2002 study questioning the use of his own work as the scientific basis for SBS/AHT. This notwithstanding the fact that the State's own expert testified that "what we know about shaking, and the established thresholds for intracranial injury comes from that study" and that Dr. Caffey based his own findings regarding SBS/AHT theory on Dr. Ommaya's 1968 study.

Based on the State's proffered evidence -- Dr. Medina's testimony and the various studies and articles submitted to the trial court -- the State has not established that there is general acceptance in the biomechanical community broadly -- beyond the six biomechanical engineers directly involved in this case. Indeed, studies suggest there is not consensus. See, e.g., L.A.H. Schiks et al., *Inflicted Head-Injury by Shaking-Trauma in Infants: The Importance of Spatiotemporal Variations of the Head's Rotation Center*, 13 *Sci. Rep.*, 2023, at 1, 1 (noting that "there is no consensus yet about whether shaking alone can result in loading and deforming an infant's anatomical structures beyond their failure thresholds and cause the . . . injuries" associated with SBS/AHT, and that "the validity of current infant shaking injury risk assessments and the injury thresholds on which these assessments are based . . . should be re-evaluated"); Mark A. Davison et al., *A Biomechanical Assessment of Shaken Baby Syndrome: What About the Spine?* 163 *World Neurosurgery* 2022, at e1,

e6 (showing that when shaking occurs, a child’s “neck w[ould] sustain injury prior to the development of closed head injury”); John Lloyd et al., Biomechanical Evaluation of Head Kinematics During Infant Shaking Versus Pediatric Activities of Daily Living, 2 J. Forensic Biomechanics, 2011, at 1, 6 (explaining that this biomechanical “study . . . like others before it, demonstrates that an adult’s shaking of an infant surrogate does not even approach the angular accelerations generally accepted as a minimum threshold for infant [development of subdural hematoma] and [diffuse axonal injury]”); Ommaya et al., 16 Brit. J. of Neurosurgery at 221, 233 (highlighting that the “experimental basis of” the theory of SBS/AHT was the Ommaya 1968 study but that those who analogized to the study in developing the theory did not realize “that the energy level of acceleration in [the study] related to speeds at motor vehicle crashes at 30 mph” and explaining that biomechanical calculations predict that “the levels of force required for retinal bleeding by shaking to damage the eye directly is biomechanically improbable”).²⁴

²⁴ The dissent states that the American Institute for Medical and Biological Engineering (AIMBE) has participated in court as amicus but “conspicuously submitted no amicus brief in this case, or any other case regarding SBS/AHT.” Post at ____ (slip op. at 38). It appears that AIMBE has participated in only one court case as amicus, the case cited by the dissent, which did not involve scientific theory, but involved diversity in higher education and the potential effect on the medical profession if race conscious admissions were banned. See Brief for Amici Curiae Association of American Medical Colleges et al. in

Nieves's experts' testimony at the Frye hearing emphasized that lack of general acceptance of SBS/AHT. Each defense expert stated that no biomechanical study has shown that shaking alone results in the symptoms associated with SBS/AHT. Dr. Van Ee testified that the scientific data does not support the proposition that someone could create enough force through shaking alone to cause a subdural hematoma, but somehow not cause injury to the infant's neck, which is very weak. Here, it bears noting that no such injury to the neck was ever identified.

The experts further testified that they believed something other than SBS/AHT could have caused D.J.'s symptoms. Dr. Scheller testified that subdural hygroma is a common finding in children born prematurely and could be the cause of D.J.'s subdural hematoma given that he was born exceptionally

Support of Respondents at 3, 33, Students for Fair Admissions, Inc. v. President & Fellows of Harvard Coll., 600 U.S. 181 (2023) (Nos. 20-1199 & 21-707). AIMBE was not the lead amicus on the brief. Ibid. The dissent also notes that AIMBE has issued no consensus statement regarding "the biomechanical community's supposed rejection of SBS/AHT." Post at ____ (slip op. at 38). AIMBE's website, however, states the following regarding consensus statements: "[AIMBE's] Council [of Societies] meets each winter at the AIMBE Annual Event, and major initiatives undertaken at recent events have included the formulation of committees to address national workforce initiatives, implementing national innovation agendas, developing consensus statements regarding federal research funding levels, and assessing the future of bioengineering through analyzing its economic impact." About AIMBE, AIMBE, <https://aimbe.org/about-aimbe/> (emphasis added) (last visited Nov. 3, 2025). As articulated by the organization itself, AIMBE's initiatives regarding consensus statements are focused on federal research funding levels.

prematurely at 25 weeks. Dr. Mack testified that the cause of D.J.'s symptoms could be the result of BESS or BEH, two conditions that result in the symptoms associated with SBS/AHT. She also testified that some of D.J.'s symptoms could be attributed to the fact that he was born very prematurely. Dr. Van Ee testified about the Prange study, which showed that short falls, at heights as low as one foot, demonstrated angular acceleration levels greater than those recorded from shaking alone, and concluded that there is no data that shaking alone can give rise to the symptoms associated with SBS/AHT.

The State and amici in support point to and argue that confession studies are strong evidence that shaking alone results in SBS/AHT. But there are several flaws with relying on confessions to establish a diagnosis within a reasonable degree of medical or scientific certainty. First, confessions are not scientific. A defendant's confession to a particular action does not equal the controlled setting of a scientific study conducted in a manner that can replicate and confirm the accuracy of results. Second, as defendants and several amici in support point out, the confession studies do not, and in some instances likely cannot, account for false confessions.

Third, partial confessions, in which the accused may give only a half truthful account of what transpired, are untrustworthy in the context of SBS/AHT studies that rely on the confessor's true and accurate account of the

alleged crime to collect scientific data. See Suzanne P. Starling et al., Analysis of Perpetrator Admissions to Inflicted Traumatic Brain Injury in Children, 158 Archives Pediatric & Adolescent Med. 454, 457 (2004) (conceding -- within a study concluding that confession data suggested that shaking without impact can result in traumatic brain injury symptoms -- that “[m]any of these admissions may contain only partial descriptions of the events, including details that may meet the minimum requirement for the legal prosecution but lack detailed descriptions necessary for biomechanical evaluation. There seems to be no correlation between the mechanism of injury admitted by perpetrators and the initial symptoms on presentation”).

Indeed, a caregiver suspect presented with the prospect of admitting to shaking and striking an infant’s head or admitting to just shaking the child may partially confess to shaking alone as opposed to shaking in conjunction with blows to the head, thus potentially never providing a full account of the events that took place. See Eyal Peer et al., “I Cheated, but Only a Little”: Partial Confessions to Unethical Behavior, 106 J. Personality & Soc. Psych. 202, 212 (2014) (“[W]hen one confesses to the full offence, one has to face up to the full consequences of admitting a major unethical act. Partial confessions thus balance the need to appear credible with the negative consequences of confessing to the full extent.”). For those reasons, it is difficult to view

medical studies based on confessions as the gold standard in determining whether the science supports an SBS/AHT diagnosis when impact is not present.

For all the foregoing reasons, we hold that the State has failed to carry its burden that Dr. Medina's expert testimony regarding SBS/AHT without impact is reliable and therefore cannot be admitted at trial.²⁵ The State has not met its burden of establishing general acceptance in the relevant scientific communities because the research, studies, and testimony presented at the hearing reflect a lack of general acceptance in the biomechanical community regarding SBS/AHT without impact. There is evidence of general acceptance by many in the medical community, but the State must also establish general acceptance in the biomechanical community, and it has failed to do so.

²⁵ The dissent claims that limiting our holding to shaking without impact fails because "shaking with impact against a soft surface, like shaking alone, may leave no external signs of trauma." Post at ____ (slip op. at 22-23). The distinction between shaking alone and shaking with impact, however, is important and often identified. In fact, the Consensus Statement, on which the dissent significantly relies, makes that very distinction itself. See Choudhary et al., 48 Pediatric Radiology at 1050 ("confessional evidence is quite striking that shaking alone can cause AHT"; "AHT can be caused by shaking alone, shaking with impact, or blunt impact alone" (emphases added)). All the witnesses at the hearing were questioned and testified at length regarding "shaking alone" and whether that could cause the symptoms associated with SBS/AHT. The distinction between shaking alone and shaking with impact is plainly the nucleus of the current debate regarding SBS/AHT.

Surely, if there is physical evidence of trauma to a child or other evidence of abuse, the State can present such evidence to a jury. And if new, reliable, scientific evidence is developed, the State can, of course, in a future case, make a showing under the Daubert standard this Court adopted in Olenowski I, that expert testimony regarding SBS/AHT without impact is reliable. In such a case, scientific evidence and research, both old and new, could be presented and considered. Science is constantly evolving, so the door is not forever closed on making such a showing of reliability.

VII. CONCLUSION

As noted, there is no dispute that child abuse is a serious and unacceptably cruel act against the most vulnerable and innocent in our society. There is no dispute that when a young child presents at a hospital or other medical facility with symptoms including subdural hematomas, retinal hemorrhages, and encephalopathy, such symptoms present a worrisome and urgent situation. No one disputes that medical professionals, including pediatricians, radiologists, neurologists, neurosurgeons, ophthalmologists, and more perform admirable work every single day to care for severely ill children who present with the symptoms associated with SBS/AHT.

The question before this Court, however, is whether the State has met its burden of proving that an expert -- Dr. Medina -- should be allowed to take the

witness stand and testify, not only about the injuries observed on a child through medical examinations and tests, but also that the only explanation for those injuries is child abuse. We hold that the State has not met that burden, and that Dr. Medina's testimony is therefore unreliable and inadmissible at trial. We affirm the Appellate Division's judgment.

CHIEF JUSTICE RABNER and JUSTICES PATTERSON, FASCIALE, NORIEGA, and HOFFMAN join in JUSTICE PIERRE-LOUIS's opinion. JUSTICE WAINER APTER filed a dissent.

State of New Jersey,
Plaintiff-Appellant,
v.
Darryl Nieves,
Defendant-Respondent.

State of New Jersey,
Plaintiff-Appellant,
v.
Michael Cifelli,
Defendant-Respondent.

JUSTICE WAINER APTER, dissenting.

This Court has previously warned that a “court should not substitute its judgment for that of the relevant scientific community.” Landrigan v. Celotex Corp., 127 N.J. 404, 414 (1992). We have cautioned that “[g]reat difficulties can arise when judges, assuming the role of scientist, attempt to assess the validity of a complex scientific methodology.” Rubanick v. Witco Chem. Corp., 125 N.J. 421, 451 (1991). Because I believe the majority fails to heed that wise counsel, I respectfully dissent.

I.

A.

As the majority explains, the Frye standard requires trial courts to determine whether the science that underlies “the proposed expert testimony has ‘gained general acceptance in the particular field in which it belongs.’” State v. J.L.G., 234 N.J. 265, 280 (2018) (quoting Frye v. United States, 293 F. 1013, 1014 (D.C. Cir. 1923)). The proponent of the evidence can establish general acceptance through expert testimony, scientific and legal writings, or judicial opinions. State v. Cassidy, 235 N.J. 482, 492 (2018).

We have long recognized that general acceptance “need not be predicated upon a unanimous belief or universal agreement in the total or absolute infallibility of the techniques, methodology or procedures that underlie the scientific evidence.” Romano v. Kimmelman, 96 N.J. 66, 80 (1984). “Although we look for wide support within the relevant scientific community, complete agreement is not required for evidence to be admitted.” J.L.G., 234 N.J. at 281. This is so because “[t]here will always be some detractors to any scientific theory.” Windmere, Inc. v. Int’l Ins. Co., 105 N.J. 373, 379 (1987); see also State v. Johnson, 905 P.2d 1002, 1010 (Ariz. Ct. App. 1995) (“If the [general acceptance] requirements were met only if there were no debate on a subject, even Copernicus’s theory of a sun-centered solar

system could not be mentioned in a court of law. The flat earth society would carry the day.” (alteration in original) (quoting People v. Soto, 35 Cal. Rptr. 2d 846, 856 (Ct. App. 1994))), aff’d, 922 P.2d 294 (Ariz. 1996).

Judicial humility requires us to accept that we are judges, not scientists. Nor are we experts in the scientific method. Courts around the country have recognized that the benefit of the Frye standard is that judges defer to scientists when assessing the scientific merit of any particular scientific theory. See, e.g., Grady v. Frito-Lay, Inc., 839 A.2d 1038, 1045 (Pa. 2003) (under the Frye standard, “the decisions of individual judges, whose backgrounds in science may vary widely, will be . . . guided by the consensus that exists in the scientific community on such matters”); Goeb v. Tharaldson, 615 N.W.2d 800, 813 (Minn. 2000) (“[A]dmissibility can be based on scientific merit only if judges defer to practicing scientists’ assessments of scientific merit.” (quotation omitted)); State v. Copeland, 922 P.2d 1304, 1312 (Wash. 1996) (“The Frye standard recognizes that ‘judges do not have the expertise required to decide whether a challenged scientific theory is correct,’ and therefore courts ‘defer this judgment to scientists.’” (quoting State v. Cauthron, 846 P.2d 502, 505 (Wash. 1993))); United States v. Addison, 498 F.2d 741, 743-44 (D.C. Cir. 1974) (“The requirement of general acceptance in the scientific

community assures that those most qualified to assess the general validity of a scientific method will have the determinative voice.”).

B.

In this case, recognizing that we are not scientists requires acknowledging that what the majority calls Shaken Baby Syndrome/Abusive Head Trauma (SBS/AHT), and what scientists simply call Abusive Head Trauma (AHT),¹ is generally accepted by those who are. Indeed, there is more evidence of general acceptance of SBS/AHT as a medical diagnosis than in most Frye cases, because SBS/AHT has been explicitly affirmed by every major discipline involved in its diagnosis and treatment. See Arabinda Kumar Choudhary et al., Consensus Statement on Abusive Head Trauma in Infants and Young Children, 48 Pediatric Radiology 1048, 1050 (2018) (Consensus Statement).

The majority references this Consensus Statement, but states that it was published by “a group of physicians and pediatric radiologists,” ante at ____

¹ I discuss below, see infra § II.A., why the majority is incorrect in characterizing this case as about “SBS/AHT without impact” or “shaking without impact.” See, e.g., ante at ____ (slip op. at 92, 97, 101). And, as described in section II.A., the medical diagnosis is Abusive Head Trauma, not “Shaken Baby Syndrome/Abusive Head Trauma.” I nonetheless use the majority’s “SBS/AHT” acronym to make clear that I am referring to the same medical diagnosis as the majority.

(slip op. at 25), implying that it was supported only by several individual doctors. It was not.

The statement was endorsed by the national and international professional societies of every major discipline involved in the diagnosis and treatment of SBS/AHT. That includes the American Academy of Pediatrics (AAP), the American Society of Pediatric Neuroradiology, the American Association for Pediatric Ophthalmology and Strabismus, the Executive Committee of the American College of Radiology, the American Professional Society on the Abuse of Children, the European Society of Neuroradiology, the Swedish Paediatric Society, the Norwegian Pediatric Association, the Japanese Pediatric Society, the Society for Pediatric Radiology, the European Society of Paediatric Radiology, the Sociedad Latino Americana de Radiología Pediátrica, the Société Francophone d’Imagerie Pédiatrique et Périnatale, the Asian and Oceanic Society for Paediatric Radiology, and the Australian and New Zealand Society for Paediatric Radiology. Consensus Statement at 1050; Arabinda Kumar Choudhary et al., Consensus Statement on Abusive Head Trauma: Additional Endorsements, 49 Pediatric Radiology 421, 421 (2019).

And it was not only endorsed by the boards of those societies. It was “vetted through a process that offer[ed] all members” of each society the opportunity to “contribute” to the statement itself. Consensus Statement at

1058. Specifically, the statement was drafted by a group of doctors. Ibid. It was then “sent to the governing body” of each medical society that ultimately signed on. Ibid. The governing bodies then “circulate[d] the document to the society membership for comment and if necessary further revisions.” Ibid. Only after this “comprehensive creation and review process,” which allowed individual members of each medical society to request changes, was the document published. Ibid.

While membership lists of each society are not included in the record, the American Academy of Pediatrics alone has 67,000 individual members. See AAP, <https://www.aap.org> (last visited Nov. 18, 2025). The Consensus Statement thus represents the views of tens of thousands, if not hundreds of thousands, of clinicians.

Those clinicians explain that abusive head trauma “is the leading cause of fatal head injuries in children younger than 2 years and is responsible for 53% of serious or fatal traumatic brain injury cases.” Consensus Statement at 1049. They state that “the medical literature and overwhelming clinical experience and judgment demonstrate that AHT can be caused by shaking alone, shaking with impact, or blunt impact alone.” Id. at 1051. They therefore use the term “[a]busive head trauma (AHT)” as the “most appropriate

and inclusive diagnostic term for infants and young children who suffer from inflicted intracranial and associated spinal injury.” Id. at 1059.

The statement affirms that “AHT is a scientifically non-controversial medical diagnosis broadly recognized and managed throughout the world.” Id. at 1049. “When diagnosed,” it means only “that accidental and disease processes cannot plausibly explain the etiology of the infant [or] child’s injuries.” Ibid.

It notes that in cases involving AHT, “defense attorneys and their retained medical witnesses have increasingly challenged longstanding medical consensus that infant shaking can cause brain trauma . . . tr[ying] to create a medical controversy where there is none.” Id. at 1058-59. However, the resulting debate “surrounding AHT is neither scientific nor medical but legal.” Id. at 1059 (quotation marks omitted). The statement explains that this “manufacture[d] . . . controversy” has turned into a “significant medical, legal and public health” concern because it has led to dissemination of the “inaccurate and dangerous message[] . . . that shaking an infant cannot cause serious injury.” Id. at 1050, 1059. That risks “encouraging dangerous or even life-threatening caregiver behavior.” Id. at 1050.

The doctors thus reiterate that a “diagnosis of AHT is made like any other medical diagnosis, by considering all the information acquired via

clinical history, physical examination, and laboratory and imaging data.” Id. at 1052. Infants suffering from AHT might present with many different neurologic signs and symptoms. Id. at 1052. Although the injuries most commonly associated with AHT include “[subdural hematoma], complex retinal hemorrhage and/or retinoschisis, rib, metaphyseal or other fractures and soft-tissue injury,” AHT is not diagnosed based only on those findings. Id. at 1059. Rather, it is a “complex and multifaceted diagnostic process” in which each infant must be “further evaluated for other diseases that might present with similar findings.” Id. at 1050, 1059. The ultimate “question to be answered is, ‘Is there a medical cause to explain the findings or did this child suffer from inflicted injury?’” Id. at 1059.

According to the Consensus Statement, “[e]fforts to create doubt about AHT” insist that it is a “near-mechanical determination based on the ‘triad’ [of symptoms] -- the findings of subdural hemorrhage, retinal hemorrhage and encephalopathy.” Id. at 1050. However, “[t]he term ‘triad’ is a legal” construct “that falsely mischaracterizes a complex AHT diagnosis process.” Id. at 1060. In reality, “AHT is a medical diagnosis . . . made only after careful consideration of all historical, clinical and laboratory findings as well

as radiologic investigations by . . . a multidisciplinary team.” Id. at 1049-50.²

The Consensus Statement concludes that although “[i]t is increasingly popular for defense lawyers to argue that AHT is a medical diagnosis of murder,” a medical expert who diagnoses a child with AHT “plays just one role -- to help the judge or jury answer the medical question of whether an infant’s injuries were most likely caused by abuse or they could be plausibly explained by a recognized disease or by . . . myriad hypothetical alternative[s].” Id. at 1059.

Beyond the societies that signed on to the Consensus Statement, AHT/SBS has also been endorsed by the World Health Organization, the American Association of Neurological Surgeons, the American Academy of

² This is confirmed by the facts of this case. D.J. was diagnosed with AHT on April 26, 2017 -- more than two months after being admitted to the hospital -- only after doctors in the neurology, genetics, hematology, and pediatric ophthalmology departments could identify no other explanation for his injuries. D.J. was originally evaluated by doctors in neurology, genetics, hematology, neuroradiology, and pediatric ophthalmology, and by a retinal specialist. He was diagnosed by a neuroradiologist with bilateral subdural bleeding. An ophthalmologist and retinal specialist diagnosed severe retinal hemorrhages in both eyes, “described as extensive, too numerous to count, involving multiple layers of the retina and extending to the periphery” and “consistent with nonaccidental trauma.” Ante at ____ (slip op. at 28, 38). D.J. then underwent a comprehensive metabolic evaluation by a geneticist to look for a metabolic condition that could cause his symptoms. After being discharged from the hospital, D.J. had follow-ups with hematology, genetics, and a retinal specialist. No metabolic or bleeding disorder was identified, and a team of doctors ruled out hyperacute increase in intracranial pressure and aneurysm. Potential accidental trauma was ruled out because D.J.’s parents denied that any occurred.

Ophthalmology, the Centers for Disease Control and Prevention, the Royal College of Ophthalmologists and the Royal College of Paediatrics and Child Health, and the French Society of Physical Medicine and Rehabilitation. See Abusive Head Trauma, Am. Acad. of Pediatrics (Nov. 18, 2025), <https://www.aap.org/en/patient-care/child-abuse-and-neglect/abusive-head-trauma>.

Similarly, in an amicus brief submitted in this case, the New Jersey State Chapter of the AAP, the American Association for Pediatric Ophthalmology and Strabismus, the American Society of Pediatric Neurosurgeons, the American Society of Pediatric Neuroradiology, the American Professional Society on the Abuse of Children, the Society for Pediatric Radiology, and the Ray E. Helfer Society -- “representing over 70,000 clinicians, treating physicians, pediatric neurologists, ophthalmologists, radiologists, and child abuse prevention experts” -- reiterate that “[t]he relevant scientific community overwhelmingly accepts the AHT diagnosis and science as valid.” They describe how “a specific pattern of retinal hemorrhages -- those that are too numerous to count, in multiple layers of the retina, and extending to the retinal periphery -- are highly specific for AHT.” And they explain that “[v]enous injury -- specifically, disruption of bridging veins where they meet the superior sagittal sinus complex (the main vein between the cerebral hemispheres) -- is considered the source of subdural hematomas and is strongly associated with

AHT.”³ They conclude that “[t]he AHT diagnosis is the result of decades of evidence-based studies repeatedly corroborated by researchers from across disciplines.”

C.

This case is therefore completely unlike the few prior instances in which we have determined that evidence did not satisfy the Frye standard. In those cases, the scientific evidence lacked the endorsement of even one major scientific association or society. Here, SBS/AHT is endorsed by all of them.

In 1982, we prohibited a defendant from offering expert testimony from a psychiatrist “who would testify that [he] d[id] not have the psychological traits of a rapist.” State v. Cavallo, 88 N.J. 508, 512 (1982). We noted that the defendant had cited nothing to suggest “that the scientific community

³ The majority highlights that when defense counsel asked Dr. Medina “there is no study that shows the tearing of the bridging vein; right?” Dr. Medina responded “no.” But that does not amount to Dr. Medina “conced[ing] that no studies show a relation between tearing of a bridging vein and subdural hematomas.” See ante at ____ (slip op. at 40). Instead, defense counsel’s follow-up questions and Dr. Medina’s answers make clear that she was explaining that she was unaware of any biomechanical study in which researchers intentionally tore a child’s bridging vein. Although, as discussed below, see infra at ____ (slip op. at 33-34), there are obvious moral and ethical reasons why such a study is not possible, the association between subdural hemorrhages and damage to bridging veins is well-documented in the medical literature, and Dr. Medina did not state otherwise. See, e.g., Consensus Statement at 1054 (“Venous injury is strongly associated with AHT. It is common at the junction of the bridging vein and superior sagittal sinus complex and is considered the source of [subdural hematoma].”).

generally accepts the existence of identifiable character traits common to rapists.” Id. at 522, 529.

We next questioned the reliability of voiceprint analysis, in which a human purported to use a machine developed at Michigan State University to “determin[e] the identity of a human voice.” Windmere, 105 N.J. at 375, 377, 380. We noted that all experts who testified to the machine’s reliability were “affiliated with the development of the device,” and one even admitted “that the device lacked scientific consensus.” Id. at 380, 382. We also emphasized that “[m]embers of the Speech Communications Section of the Acoustical Society of America voted 42-0 against the reliability of the procedure.” Id. at 384.

In State v. Harvey, we rejected the State’s claim that a professor was able to estimate a person’s height “from the size of his or her shoes.” 121 N.J. 407, 426 (1990). We found that the State had not proven general acceptance of the theory because “[i]t did not provide evidence that anyone in the scientific community other than [the professor] himself vouches for his methods.” Id. at 428. We also noted that we had found no other cases “in which an expert’s testimony involved the scientific comparison of sneaker prints with stature.” Ibid.

In State v. Moore, we barred “hypnotically refreshed testimony,” holding that “[t]he theory that hypnosis is a reliable means of improving recall is not generally accepted in the scientific community.” 188 N.J. 182, 210 (2006). We highlighted that 26 other states had agreed that “hypnotically refreshed testimony is not generally accepted science.” Ibid.

In 2018, we held that “Child Sexual Abuse Accommodation Syndrome” was not generally accepted. J.L.G., 234 N.J. at 271, 308. We explained that the professed psychological syndrome had been hypothesized by a clinical psychiatrist in 1983, and in the 35 years since, had never been recognized by the American Psychiatric Association or the American Psychological Association. Id. at 271-72, 281. It was also never included in “the Diagnostic and Statistical Manual of Mental Disorders (DSM-5), the mental health field’s authoritative list of mental disorders.” Id. at 272.

Finally, in State v. Cassidy, we held there was no general acceptance of an Alcotest machine calibrated without a National Institute of Standards and Technology (NIST)-traceable digital thermometer. 235 N.J. at 486-87, 497. In upholding the Special Adjudicator’s conclusion, we noted that he “found it ‘extremely important and persuasive’ that . . . protocol treats the failure to achieve an in-range temperature reading using the NIST-traceable thermometer” as requiring that the calibration be “abort[ed].” Id. at 495. We

also adopted the Special Adjudicator’s finding that other states did not “reveal[] general acceptance of the reliability of Alcotest results without the use of a NIST-traceable thermometer.” Id. at 496.

D.

The majority states that “[t]here is, of course, . . . case law finding SBS/AHT evidence reliable and therefore admissible at trial.” Ante at ____ (slip op. at 90). That is quite the understatement. Every other state admits expert testimony diagnosing SBS/AHT. And every other court that has considered the question has held that such evidence is admissible.

The cases cited by the majority, see ante at ____ (slip op. at 90-91), do not scratch the surface. Some additional cases include State v. Stewart, 923 N.W.2d 668, 676 (Minn. Ct. App. 2019) (holding that “[t]he record amply establishes that the theory of abusive head trauma is reliable”); Sissoko v. State, 182 A.3d 874, 898, 906 (Md. Ct. Spec. App. 2018) (rejecting the “appellant’s argument that the controversy over the diagnosis of abusive head trauma makes it no longer a generally accepted diagnosis in the absence of external [evidence of injury]” because “[t]hat controversy exists largely in the legal community, not in the medical communities relevant to our inquiry,” and concluding “that the diagnosis of abusive head trauma remains generally accepted in the relevant medical/scientific communities”); State v. West, 551

S.W.3d 506, 516-17 (Mo. Ct. App. 2018) (holding that there was no error in the trial court “failing to sua sponte prevent or strike testimony regarding ‘shaken baby syndrome’ and ‘abusive head trauma or injury’” because “[t]he proper way to challenge an expert’s method of reaching conclusions is through cross-examination” (quotation omitted)); In re Morris, 355 P.3d 355, 360 (Wash. Ct. App. 2015) (holding that expert “testimony satisfied the Frye standard” because “[a]busive head trauma as a diagnosis, and shaking as a cause of such injuries, are generally accepted theories in the relevant scientific community”); State v. McClary, 541 A.2d 96, 102 (Conn. 1988) (addressing “whether ‘shaken baby syndrome’ . . . is a ‘generally accepted’ diagnosis in the medical field” under Frye and “conclud[ing] that it is”).

No case has ever concluded that evidence of SBS/AHT is unreliable. And no case has ever found its reliability sufficiently questioned to preclude its admission at a civil or criminal trial. Instead, the cases the majority cites generally remanded for new trials to allow the jury to hear from experts that both supported and opposed the diagnosis of SBS/AHT. See People v. Lemons, 22 N.W.3d 42, 60-62 (Mich. 2024) (reversing conviction and remanding for a new trial where trial court precluded Dr. Chris Alan Van Ee, the defense expert in this case, from testifying, because “divergences” in expert opinion about SBS/AHT “are a matter of weight, not admissibility,” and

observing that “[o]n retrial,” the prosecution could “call the various experts . . . to testify about the validity of the SBS diagnosis and their beliefs that [the child’s] injuries were consistent” with it); Ex parte Roark, 707 S.W.3d 157, 185, 187-88 (Tex. Crim. App. 2024) (vacating conviction and remanding for a new trial because “scientific knowledge has evolved regarding SBS and its application,” and the jury should “hear experts on both sides arguing their position”); Smith v. State, 882 S.E.2d 300, 311-12 (Ga. 2022) (determining that a disagreement between the defense and prosecution about SBS/AHT “presents a factual dispute,” and remanding for an evidentiary hearing); Allison v. State, 448 P.3d 266, 273-74 (Alaska Ct. App. 2019) (reversing conviction and remanding for a new trial because the trial court erroneously excluded “all mention” of a pre-existing condition that supported the defense theory that SBS/AHT was not the cause of death, and “[e]vidence that there were other possible medical explanations . . . was something that the jury should have heard”); Commonwealth v. Millien, 50 N.E.3d 808, 809-10, 826 (Mass. 2016) (finding defense counsel ineffective for not hiring an expert to question the State’s evidence of SBS/AHT because “there is a vigorous debate on this subject, . . . arguments are being made on both sides with support in the scientific and medical literature, . . . [and] the jury heard only one side of this debate”); Commonwealth v. Epps, 53 N.E.3d 1247, 1250, 1268 (Mass. 2016)

(vacating conviction and remanding for a new trial because defense counsel called no expert to challenge SBS/AHT testimony); People v. Bailey, 999 N.Y.S.2d 713, 726-27 (N.Y. Cnty. Ct. 2014) (vacating conviction and ordering a new trial because developments in SBS/AHT research constituted “newly discovered evidence”), aff’d, 41 N.Y.S.3d 625 (N.Y. App. Div. 2016); People v. Ackley, 870 N.W.2d 858, 860, 867 (Mich. 2015) (vacating and remanding for a new trial when defense counsel “called no expert in support of its theory that the child’s injuries resulted from an accidental fall” rather than SBS/AHT, despite receiving court funding for same); Del Prete v. Thompson, 10 F. Supp. 3d 907, 909, 952, 958 (N.D. Ill. 2014) (holding defendant’s ineffective assistance of counsel claim was not procedurally barred when his trial counsel did not challenge the admissibility of the State’s SBS/AHT expert testimony); Cavazos v. Smith, 565 U.S. 1, 3-6, 8-9 (2011) (summarily reversing the Ninth Circuit’s grant of habeas relief because the jury’s guilty verdict was “supported by the record,” including SBS/AHT evidence); State v. Edmunds, 746 N.W.2d 590, 591, 599 (Wis. Ct. App. 2008) (reversing the denial of a motion for a new trial and remanding so that “a jury [c]ould be faced with competing credible medical opinions” on the reliability of SBS/AHT evidence).

As these citations reflect, some cases have acknowledged a debate around SBS/AHT and either allowed evidence critical of it to be admitted at trial or remanded for a new trial in which experts both for and against could testify. Until today, no court has ever held that evidence in support of SBS/AHT cannot be admitted at trial.

II.

This more than suffices to show that SBS/AHT is generally accepted in every major discipline involved in its diagnosis and treatment: pediatrics, child abuse pediatrics, neurology, neuroradiology, neurosurgery, radiology, ophthalmology, and emergency medicine. See, e.g., Sissoko, 182 A.3d at 904 (“[T]he opinion that abusive head trauma is a legitimate and reliably-made diagnosis is shared by the overwhelming majority of physicians within the relevant medical fields.”).

Yet the majority holds otherwise, concluding that “expert testimony regarding SBS/AHT without impact” is not reliable and “therefore cannot be admitted at trial” because there is a “lack of general acceptance in the biomechanical community.” Ante at ____ (slip op. at 107). There are serious flaws with the majority’s reasoning and conclusion. I address each in turn.

A.

To start, the majority repeatedly states that this case is about whether expert testimony regarding the medical diagnosis of “SBS/AHT without impact is sufficiently reliable for admission at trial.” Ante at ____ (slip op. at 94) (emphasis added); ante at ____ (slip op. at 101) (“[T]here is no general acceptance in the biomechanical community of the SBS/AHT without impact diagnosis.” (emphasis added)).

But there is no such thing. There is simply no medical diagnosis of “SBS/AHT without impact” or SBS/AHT based on “shaking alone.” That is why Dr. Gladibel Medina diagnosed D.J. with “abusive head trauma, as it occurs with a shaking event with or without impact.” See ante at ____ (slip op. at 41).

When an infant is shaken and then slammed into a soft surface, such as a crib mattress or bed, there may be no external sign of impact, such as a cut or bruise. But that does not mean the infant was only shaken, rather than shaken and slammed into a soft surface. Dr. Medina testified to this very clearly:

[Prosecutor]: And your diagnosis in this case is that there was abusive head trauma with shaking with or without impact, correct?

[Dr. Medina]: Yes.

[Prosecutor]: And when we’re talking about impact, what kind of impact are we talking about?

[Dr. Medina]: Oh. It would have to be an impact into a soft surface, because he doesn't have any external signs of trauma.

[Prosecutor]: Okay. And can you have shaking alone with no external signs of trauma?

[Dr. Medina]: Yes.

[Prosecutor]: And can you have shaking with impact with no external signs of trauma?

[Dr. Medina]: Oh, yes.

Often, when there is no external sign of impact, there is no way to know whether the infant was only shaken, or shaken and slammed into a soft surface, unless the incident was witnessed or videotaped. See, e.g., C.Z. Cory & M.D. Jones, Can Shaking Cause Fatal Brain Injury? A Biomechanical Assessment of the Duhaime Shaken Baby Syndrome Model, 43 Med., Sci. & L. 317, 319 (2003) (“children with intracranial injuries but without detectable signs of external head trauma may have suffered an impact with a padded surface, such as a cushion or crib mattress” that would not be detectable unless the incident had been videotaped); see also John W. Finnie et al., Neuropathological Changes in a Lamb Model of Non-accidental Head Injury (the Shaken Baby Syndrome), 19 J. Clinical Neurosci. 1159, 1164 (2012) (“The absence of any external evidence of head trauma also does not necessarily negate a diagnosis of [non-accidental head injury] for, if the head is impacted against a soft

surface, substantial brain damage may still be sustained from rapid angular deceleration of the head.”).

The majority emphasizes that in Dr. Ann-Christine Duhaime’s retrospective analysis of all SBS/AHT cases at the Children’s Hospital of Philadelphia over a seven-year period, “only one involved shaking without impact” and all 13 infants who died “had evidence of blunt head trauma.”

Ante at ____ (slip op. at 19) (quoting Ann-Christine Duhaime et al., The Shaken Baby Syndrome: A Clinical, Pathological, & Biomechanical Study, 66 J. Neurosurgery 409, 411 (1987)). But it does not appreciate the import of the fact that in seven of those 13 cases, “no evidence of external trauma was noted on the initial physical examination” and “impact findings were noted only at autopsy, and had not been apparent prior to death.” Duhaime, 66 J. Neurosurgery at 411, 414 (emphasis added). In other words, even for the 13 infants who were so seriously injured that they died from SBS/AHT, for seven of them, there was no evidence of “impact” while they were alive.

That hurts rather than helps the majority’s conclusion, because as even defense expert Dr. Van Ee agreed, evidence of impact that can only be seen on autopsy will “never [be] seen” when a child survives shaking, because “obviously, you’re not going to do an autopsy on a living child.”

The majority repeatedly conflates “shaking without external signs of impact” with “shaking without impact” even on a soft surface, concluding, for example, that because Dr. Medina “testified on cross-examination . . . that D.J. did not ‘have any external signs of impact,’” “Nieves’s case is properly assessed as one involving shaking without . . . impact.” Ante at ____ (slip op. at 95).

This impacts the majority’s entire analysis. Although “events” of “shaking with or without impact” may be “very different” as a factual matter, ante at ____ (slip op. at 95), unless the event is witnessed or videotaped, doctors often cannot know which occurred. See, e.g., Sissoko, 182 A.3d at 876, 906 (stating that “[t]he State’s theory of prosecution was that the appellant killed his son by violently shaking him, slamming him against a soft surface, or both” and that “[n]one of the State’s experts were able to testify conclusively whether shaking alone, impact on a soft surface alone, or a combination of both mechanisms caused the injuries,” because the incident was not witnessed or videotaped); Consensus Statement at 1052-53 (“The absence of external trauma to the head and neck is common” even in cases of shaking with impact on a soft surface because “sometimes soft-tissue injuries including scalp hematomas are only evident at autopsy.”).

Because shaking with impact against a soft surface, like shaking alone, may leave no external signs of trauma, the majority's effort to limit its holding to "expert testimony regarding SBS/AHT without impact," see, e.g., ante at ____ (slip op. at 107), fails.

B.

The majority presents a history of the "origins and evolution of SBS/AHT" that begins with Dr. Ayub K. Ommaya's 1968 experiment, ante at ____ (slip op. at 9-26) (citing Ayub K. Ommaya et al., Whiplash Injury & Brain Damage: An Experimental Study, 204 J. Am. Med. Ass'n 285, 285 (1968)), to prove that the "foundation of SBS/AHT lies in biomechanical science and engineering," ante at ____ (slip op. at 92). But no expert testified to that history. And I have found it in no publication.

Instead, Dr. Medina testified that AHT has been recognized in the medical community for 160 years, beginning when a French pathologist identified "injuries in children that were believed to be associated with maltreatment by caregivers." Her account is supported by the medical literature, which discussed AHT long before Dr. Ommaya's study or the advent of the modern field of biomechanical engineering.

In 1860, Dr. Auguste Ambroise Tardieu wrote about cases of "infanticide, including cases without external signs of injury, but where [there

had been] hemorrhage in the brain and collections of blood over the brain.”

Sandeep Narang, A Daubert Analysis of Abusive Head Trauma/Shaken Baby Syndrome, 11 Hous. J. Health L. & Pol’y 505, 523-24 (2011) (citing Auguste Ambroise Tardieu, Étude médico-légale sur les sévices et mauvais traitements exercés sur des enfants, 13 Annales d’hygiène publique et de médecine légale 361-98 (1860)). In the late 1800s, doctors theorized that those subdural hemorrhages were caused by infection. Id. at 524. But in 1914, British neurosurgeon Dr. Wilfred Trotter wrote that they were actually caused by trauma “coming from veins torn in their course between the brain and a dural sinus.” Id. at 525.

In 1946, Dr. John Caffey wrote about a “correlation in infants between long bone fractures, which often are associated with abuse, and subdural hematomas.” Id. at 505 n.1, 526 (citing John Caffey, Multiple Fractures in the Long Bones of Infants Suffering from Chronic Subdural Hematoma, 56 Am. J. Roentgenology 163-73 (1946)); Sissoko, 182 A.3d at 898. In “several of the cases,” retinal hemorrhages were also involved. Sissoko, 182 A.3d at 898. “In the two decades following Caffey’s historic article, multiple articles . . . confirmed the association of [subdural hemorrhages] with inflicted trauma” in infants. Narang, 11 Hous. J. Health L. & Pol’y at 526-27.

Then, in 1962, “eminent pediatrician” Dr. C. Henry Kempe’s “landmark article, The Battered-Child Syndrome,” explained that child abuse should be considered for any child presenting with subdural hematoma. Id. at 527 (quoting C. Henry Kempe et al., The Battered-Child Syndrome, 181 J. Am. Med. Ass’n 17, 17 (1962)).

The majority’s history includes none of this, all of which occurred before Dr. Ommaya’s 1968 monkey experiment -- instead asserting that the experiment was the “starting point” of SBS/AHT. Ante at ____ (slip op. at 92). The majority then discusses Dr. Ommaya’s study at length, referencing him 53 times. But only the majority, not any publication or the doctors who diagnose SBS/AHT, calls Dr. Ommaya’s study “the foundation of SBS/AHT theory and diagnosis in the first instance.” Ante at ____ (slip op. at 96).

Dr. Medina did not testify that Dr. Ommaya’s study was the “starting point” of SBS/AHT. Instead, she discussed it in relation to biomechanical “controversy” over whether “shaking, which has been established to cause injury in primates . . . can cause the forces needed to generate intracranial injury in infants.” And while the majority thrice quotes Dr. Medina’s testimony that “what we know about shaking, and the established thresholds for intracranial injury comes from that study” and that “everything else in biomechanics is based on those injury thresholds,” ante at ____ (slip op. at 31,

100, 102), Dr. Medina made clear that the injury thresholds in Dr. Ommaya's study were for primates -- specifically monkeys -- not human infants. She also testified that "scaling down" injury thresholds "from primates to adult human brains" and from "human brains to infant brains," "ha[s] not been validated" because "infant brains are significantly different than adult brains." Her testimony is fully supported by the medical literature, discussed further below, see infra at ____ (slip op. at 29-30).

The majority also emphasizes Dr. Duhaime's 1987 biomechanical study using what it calls "infant models," and her conclusion that "shaking alone does not produce the shaken baby syndrome," ante at ____ (slip op. at 20) (quoting Duhaime, 66 J. Neurosurgery at 409).

But the majority does not acknowledge that the "infant models" in Dr. Duhaime's study were "Just Born dolls" whose empty plastic heads were filled with cotton and water until they weighed approximately as much as an infant's head. Duhaime, 66 J. Neurosurgery at 411-12. It does not mention that one of Dr. Duhaime's models had a "hinge neck" that allowed only "anteroposterior angulation of the head" during shaking. Id. at 412. This means that, when shaken, the doll's head could only tilt backward and forward along a straight line; it could not tilt from side to side or turn in any direction. As later researchers pointed out, these models were "very simplistic and w[ere] not

designed to resemble a human infant in terms of mechanical response (biofidelity).” Cory, 43 Med., Sci. & L. at 331.

And the majority does not disclose that when Dr. Duhaime collaborated with Dr. Michael T. Prange to conduct a follow-up study, this time with the La Baby doll, they admitted that a significant limitation of both the original and follow-up studies was “the inability of the [doll’s] neck to mimic the properties of a real infant’s neck” because, “[a]t present, no detailed quantitative information is available to validate the biomechanical properties of the human infant neck.” Michael T. Prange, Brittany Coats, Ann-Christine Duhaime & Susan S. Margulies, Anthropomorphic Simulations of Falls, Shakes, and Inflicted Impacts in Infants, 99 J. Neurosurgery 143, 144, 147 (2003). To put it plainly: researchers did not know how to make a doll with a neck that resembled a human infant’s. Id. at 144.

The researchers also acknowledged that “the cervical spine consists of a series of vertebrae, allowing for rotation of the neck in different locations and directions; the hinge used to model the neck motion in the dummy [in both the original and follow up experiments] has a fixed point of rotation, only allowing anteroposterior flexion and extension.” Id. at 148. In other words, again, while in real life violent shaking can cause an infant’s head to move in

many ways, in these experiments, the doll's head could only move forward and back in a straight line. Ibid.⁴

It is therefore unsurprising that when later researchers attempted to reconstruct Dr. Duhaime's original experiment, they discovered not only that it had substantial flaws, but also that making small changes to the doll led to drastically different results. Dr. C.Z. Cory, for example, discovered that changing the doll's "centre of gravity, neck type and chest and back padding" from cotton to cotton and wool led to higher angular accelerations than Dr. Duhaime had found and surpassed the injury thresholds that Dr. Duhaime had used for concussions. Cory, 43 Med., Sci. & L. at 329. Dr. Cory concluded that "changes in the biomechanical properties of the model influence the results for angular head accelerations. Neglecting these factors produces a model with an unknown resemblance to an infant, therefore, any simulation results obtained with such a model will be meaningless and conclusions drawn unreliable." Id. at 331 (emphasis added). Dr. Cory thus determined: "[I]t

⁴ The majority repeatedly references Dr. Van Ee's testimony, based on the Prange study, that angular accelerations of a baby falling over and hitting its head on carpet, even from "heights as low as one foot," were greater than angular accelerations from shaking, and there is therefore "no data that shaking alone can give rise to the symptoms associated with SBS/AHT." See, e.g., ante at ____ (slip op. at 54-55, 105). However, the majority ignores this obvious defect with the shaking performed in the Prange study.

cannot be categorically stated, from the [Duhaime] study, that ‘pure shaking’ cannot cause fatal head injuries in an infant.” Id. at 332.

In 2017, Dr. Carole A. Jenny conducted experiments with “a customized 12-segment, instrumented infant [anthropomorphic test device] that represents a 5th percentile Japanese newborn.” Carole A. Jenny et al., Biomechanical Response of the Infant Head to Shaking: An Experimental Investigation, 34 J. Neurotrauma 1, 1-2 (2017). Whereas the Prange torso was made of wood, with no arms or legs, Dr. Jenny’s doll had “a segmented flexible torso/spine,” along with a head, neck, arms, and legs. Id. at 5. And whereas the dolls used by Drs. Duhaime and Prange had hinged necks that “constrained head-neck motions to the sagittal plane (anterior-posterior directions),” Jenny’s “neck was free to move in three planes . . . the sagittal, coronal, and transverse planes.” Id. at 5-6. In other words, while the hinged-neck doll heads used by Drs. Prange and Duhaime could only tilt forward and back, the doll head in Dr. Jenny’s study could tilt forward and back, tilt side to side, and turn right to left. It is thus unsurprising that Dr. Jenny’s “maximum angular head acceleration was . . . 13,260 rad/sec,” a “10-fold increase over Duhaime’s findings.” Id. at 4 (emphasis added).

The majority likewise does not acknowledge what all of these biomechanical studies do: because we, thankfully, do not intentionally shake

infants for research, scientists do not know what level of acceleration or velocity is required to injure an infant's brain or neck. In 2003, Dr. Prange explicitly stated that "the results of the dummy tests cannot be used to predict whether . . . rotations are sufficient to cause injury" because injury "thresholds are currently unavailable for the pediatric population." Prange, 99 J. Neurosurgery at 148. Dr. Cory agreed, noting that while Dr. Duhaime and her colleagues attempted to "scale" injury thresholds from "subhuman primates," and other studies attempted to scale from human adults, there were significant "doubts about the validity" of both. Cory, 43 Med., Sci. & L. at 327, 329. Even in 2017, Dr. Jenny emphasized that there were still "no validated infant brain injury thresholds." Jenny, 34 J. Neurotrauma at 8. She also stated that attempting to scale injury thresholds "from adult cadaver and primate studies to infants based on brain mass" was dangerous because "[o]ther factors, such as brain material properties and geometry, must also be considered when attempting to assess brain injury risk." Ibid.

In short, there are no accepted thresholds for infant brain or neck injury. The majority's assertion that there is "no test supporting a finding that humans can produce the physical force necessary to cause the symptoms associated

with SBS/AHT in a child,” see, e.g., ante at ____ (slip op. at 99), disregards that it is unknown what amount of force that would be.⁵

The majority also discounts the importance of Dr. John W. Finnie’s 2012 experiment in which nine anesthetized lambs were “vigorously shaken . . . similar to head motions believed to occur” in SBS/AHT. Finnie, 19 J. Clinical Neurosci. at 1159. The researchers’ plan was to kill the lambs six hours post-shaking and then study their brains, cervical spinal cords, and eyes. Id. at 1160. However, three of the smaller lambs “died unexpectedly at two, three and five hours after the last episode of shaking.” Ibid. Those lambs had diffuse axonal injury in their brains, including in the brainstem and at the craniocervical junction (where the head meets the upper spine). Id. at 1163. They also suffered “multifocal damage” to the retina. Id. at 1162, 1164.⁶

⁵ This similarly negates Dr. Joseph Scheller’s testimony that there have been no biomechanical studies “that have been able to produce the amount of force needed to cause a baby to suffer a subdural hematoma with shaking alone,” ante at ____ (slip op. at 43), and Dr. Van Ee’s testimony that biomechanics “does not support the proposition that someone could create enough force through shaking alone to cause a subdural hematoma, but somehow not cause injury to the infant’s neck,” ante at ____ (slip op. at 104).

⁶ Dr. Finnie’s 2012 study thus contradicts Dr. Van Ee’s testimony that no animal studies have ever “proved that subdural hematomas and retinal hemorrhages can be caused by shaking alone,” ante at ____ (slip op. at 55-56), and testimony of the defense experts “that no biomechanical study has shown that shaking alone results in the symptoms associated with SBS/AHT,” ante at ____ (slip op. at 104).

The study observed: “While this is an animal model . . . the lamb nevertheless resembles a human infant in important respects . . . namely having a relatively large gyrencephalic brain and weak neck muscles supporting the head.” Ibid. The authors therefore concluded: “[O]ur finding that shaking alone resulted in death . . . is evidence that a head impact is not always needed” to cause death in cases of AHT. Id. at 1164.

The majority does not explain why it accords more weight to Drs. Duhaime and Prange’s studies of hinged-neck plastic dolls than to Dr. Finnie’s finding that shaking alone caused death in animals that share relevant anatomical traits with human infants.

C.

The majority rejects scientific support for SBS/AHT only by imposing requirements that apply to no other diagnoses.

First, the majority diminishes studies in which defendants confessed to shaking because “[a] defendant’s confession to a particular action does not equal the controlled setting of a scientific study conducted in a manner that can replicate and confirm the accuracy of results.” Ante at ____ (slip op. at 105); see also ante at ____ (slip op. at 106-07) (“[I]t is difficult to view medical studies based on confessions as the gold standard in determining whether the

science supports an SBS/AHT diagnosis when impact is not present.”).⁷ And the majority repeatedly emphasizes that there are no randomized controlled trials of SBS/AHT. Ante at ____ (slip op. at 58) (quoting the trial court’s statement that “no one has ever tested the capacity of an individual to shake a baby in an effort to cause the triad of symptoms”); ante at ____ (slip op. at 12-14, 18 (criticizing Dr. Caffey for not “conduct[ing] any biomechanical experiments himself”); ante at ____ (slip op. at 12) (criticizing Dr. Arthur Norman Guthkelch for same).

For moral and ethical reasons, it is not possible to conduct randomized controlled trials of what happens when an adult intentionally shakes a live infant. See, e.g., Lemons, 22 N.W.3d at 57 (“[T]here can never be a perfectly replicated model of a shaken infant for obvious ethical reasons.”); Wolfe v. State, 509 S.W.3d 325, 339 (Tex. Crim. App. 2017) (“[T]here is no way to engage in controlled trials to directly test the degree of force required to cause

⁷ The majority also recounts Dr. Scheller’s testimony that “there have been no reports of witnessed shaking where the result was subdural hematomas, retinal hemorrhages, or neck injury,” ante at ____ (slip op. at 43), and no “study that shows a human can shake a baby causing the triad of injuries,” ante at ____ (slip op. at 48). That is directly contradicted by a study in the record that documented children who developed subdural and retinal hemorrhages after witnessed shaking. See Kenneth W. Feldman et al., Abusive Head Trauma Follows Witnessed Infant Shaking, 31 Child Abuse Rev. e2739, at 1 (2022) (concluding that the study “provide[s] support for isolated shaking as a cause of AHT”).

these injuries in an infant”). Indeed, even the five amici biomechanical engineers note that “the response of living human tissue most often cannot be studied ethically in possibly injurious situations.”

In any event, “[a]lmost all well-established, undisputed medical diagnoses have no randomized controlled trials (RCTs) supporting or validating their diagnostic criteria.” Narang, 11 Hous. J. Health L. & Pol’y at 532. For example, although migraine headaches are a valid medical diagnosis, and there is substantial medical literature on their evaluation, diagnosis, and treatment, “there is not one RCT evaluating the diagnostic criteria for migraine headaches, or their validity.” Ibid. The same is true for the common cold, ear infections, and all psychiatric disorders. Id. at 533. “In short, the requirement that an RCT is necessary in order to validate diagnostic criteria of a particular medical diagnosis is not only inaccurate but also inconsistent with the vast majority of clinical medicine.” Ibid.

Second, the majority highlights the trial court’s conclusion that SBS/AHT “is more conjecture than a diagnosis because it is an option embraced once a diagnostician runs out of diagnostic options,” and is “a final option lacking a reliable diagnostic criteria masking as a diagnosis.” Ante at ____ (slip op. at 57, 60). And it underscores Dr. Julie Mack’s testimony that there is “no confirmatory test” for SBS/AHT. Ante at ____ (slip op. at 49).

This means that SBS/AHT is a differential diagnosis, which we have previously defined as “a patient-specific process of elimination that medical practitioners use to identify the ‘most likely’ cause of a set of signs and symptoms from a list of possible causes.” Creanga v. Jarda, 185 N.J. 345, 355 (2005) (quoting Ruggiero v. Warner-Lambert Co., 424 F.3d 249, 254 (2d Cir. 2005)).

At the first step of a differential diagnosis, the expert “‘rule[s] in’ all plausible causes for the patient’s condition by compiling ‘a comprehensive list of hypotheses that might explain the set of salient clinical findings under consideration.’” Id. at 356 (quoting Clausen v. M/V New Carissa, 339 F.3d 1049, 1057-58 (9th Cir. 2003)). At the second step, the expert “‘rule[s] out those causes that did not produce the patient’s condition by engaging ‘in a process of elimination, eliminating hypotheses on the basis of a continuing examination of the evidence so as to reach a conclusion as to the most likely cause of the findings in that particular case.’” Ibid. (quoting Clausen, 339 F.3d at 1058); see also Westberry v. Gislaved Gummi AB, 178 F.3d 257, 262 (4th Cir. 1999) (“A reliable differential diagnosis typically . . . is performed after physical examinations, the taking of medical histories, and the review of clinical tests, including laboratory tests, and generally is accomplished by determining the possible causes for the patient’s symptoms and then

eliminating each of these potential causes until reaching one that cannot be ruled out or determining which of those that cannot be excluded is the most likely.” (internal quotation omitted)).

We have acknowledged “the widespread acceptance of differential diagnosis in the medical community, the recognition of the technique in state and federal courts, and its compatibility with our rules of evidence and prior case law.” Creanga, 185 N.J. at 357. We have therefore held differential diagnoses admissible in our courts. Ibid.

And, of course, many well-established medical diagnoses are differential diagnoses, lacking a specific “confirmatory test.” See, e.g., R.A. Murden, The Diagnosis of Alzheimer’s Disease, in 282 Advances in Experimental Medicine and Biology 59, 59 (T. Zandi & R.J. Ham eds., 1990) (“The diagnosis of Alzheimer’s Disease has traditionally been a diagnosis of exclusion.”); E. Capelli et al., Chronic Fatigue Syndrome/Myalgic Encephalomyelitis: An Update, 23 Int’l J. Immunopathology & Pharmacology 981, 981 (2010) (“A diagnosis of [Chronic Fatigue Syndrome] is a diagnosis of exclusion . . .”). Why that is generally accepted across medicine, but unreliable when it comes to SBS/AHT, the majority does not explain.

D.

The majority ultimately concludes that “[t]he State has not met its burden of establishing general acceptance in the relevant scientific communities” because, although “[t]here is evidence of general acceptance by many in the medical community, . . . the State must also establish general acceptance in the biomechanical community, and it has failed to do so.” Ante at ____ (slip op. at 107).

But the majority’s conclusion about the “biomechanical community” is based not on any consensus statement by a society of biomechanical engineers. It is based on the testimony of one biomechanical engineer (Dr. Van Ee),⁸ an amicus brief submitted by five additional biomechanical engineers (Lindsay “Dutch” Johnson, Ph.D.; Ken Monson, Ph.D.; Kirk Thibault, Ph.D., D-IBFES; Keith Button, Ph.D., PE; and Johan Ivarsson, Ph.D.), and the majority’s views of studies published in biomechanical journals or conducted by individual biomechanists.

Associations of biomechanical engineers do exist. For example, the American Institute for Medical and Biological Engineering (AIMBE) comprises 2,000 members of “outstanding” merit and reaches a broader

⁸ Dr. Van Ee stated that he has testified for the defense in SBS/AHT cases at least 70 to 80 times.

community of 50,000 people. About AIMBE, AIMBE, <https://aimbe.org/about-aimbe/> (last visited Nov. 18, 2025). It has put out no consensus statement around the biomechanical community’s supposed rejection of SBS/AHT.

AIMBE has also participated in court as amicus curiae. See Brief for Amici Curiae Association of American Medical Colleges et al. in Support of Respondents at 3, 33, Students for Fair Admissions, Inc. v. President & Fellows of Harvard Coll., 600 U.S. 181 (2023) (Nos. 20-1199 & 21-707) (arguing that “[d]iversity in the education of the Nation’s . . . healthcare professionals is a medical imperative” and that “[b]anning race-conscious admissions . . . will imperil the health and lives of Americans”). Yet it has conspicuously submitted no amicus brief in this case, or any other case regarding SBS/AHT.

Relying on the individual views of six biomechanical engineers and specific biomechanical studies poses real risks of selection bias and false equivalence. As the Consensus Statement highlights, attempting to measure general acceptance based on the testimony or statements of individuals leaves courts susceptible to “fringe, speculative, or professionally irresponsible opinions.” Consensus Statement at 1058. On the other hand, consensus statements that are “vetted by the membership” of endorsing societies and

organizations provide courts with true information about what is generally accepted in a particular field. Ibid. And some disagreement, which is present in every scientific field, does not justify the conclusion that there is no general acceptance in a particular community. See, e.g., State v. Johnson, 42 N.J. 146, 171 (1964) (“Practically every new scientific discovery has its detractors and unbelievers, but neither unanimity of opinion nor universal infallibility is required.”).

More fundamentally, I disagree with the majority using a requirement of “cross-disciplinary validation,” ante at ____ (slip op. at 81) (quoting State v. Pickett, 466 N.J. Super. 270, 323 (App. Div. 2021)), to allow individual biomechanical engineers to override nearly all pediatricians, pediatric neurologists, neurosurgeons, neuroradiologists, ophthalmologists, radiologists, and child abuse pediatricians involved in the diagnosis and treatment of SBS/AHT.⁹

⁹ As the majority discusses, the defense also presented testimony from Dr. Scheller, a pediatric neurologist in private practice, who stated that he has testified for the defense in AHT cases more than 200 times. In several cases, including very recently, his testimony was deemed inadmissible because it was not “scientifically valid and reliable.” See, e.g., LeFebvre v. State, 313 A.3d 1156, 1161, 1169 (R.I. 2024). Defendants also presented testimony from Dr. Mack, who serves as the Chief of the Division of Breast Imaging at the Penn State Milton S. Hershey Medical Center and stated that she has testified for defendants in AHT cases 32 to 35 times. And defendants were supported by an amicus brief from a total of 19 individual medical doctors, one of whom is Dr. Scheller. See ante at ____ (slip op. at 68 n.19, 76).

This approach finds no footing in our law. In Pickett, the question was whether the “defendant [was] entitled to trade secrets of a private company for the sole purpose of challenging at a Frye hearing the reliability of the science underlying novel DNA analysis software.” 466 N.J. Super. at 276-77 (footnote omitted). The Appellate Division held that “if the State chooses to utilize an expert who relies on novel probabilistic genotyping software to render DNA testimony, then defendant is entitled to access, under an appropriate protective order, to the software’s source code” to “independently test whether the evidentiary software operates as intended.” Id. at 277, 279. Although several amici argued that the software “integrates multiple scientific disciplines, therefore requiring cross-disciplinary validation to determine reliability,” id. at 323, the court did not hold that disagreement from individual computer scientists would bar the State from introducing the testimony. It simply remanded to the trial court to allow the defendant to access the source code and then continue a Frye hearing to determine reliability. Id. at 324.

In this case, as the majority correctly notes, D.J. was evaluated by “doctors with expertise in neurology, genetics, hematology, neuroradiology, and pediatric ophthalmology, as well as a retinal specialist.” Ante at ____ (slip op. at 28). Of those many specialties involved in the diagnosis and treatment of SBS/AHT, each has its own area of expertise. They belong to discrete

professional bodies, many of which contributed to the Consensus Statement and to the amicus brief of the medical societies submitted in this case.

Yet the majority lumps pediatrics, child abuse pediatrics, neurology, neuroradiology, neurosurgery, radiology, ophthalmology, hematology, genetics, and emergency medicine into a catch-all “medical/pediatric community” and allows the “biomechanical engineering community” to overrule them. Ante at ____ (slip op. at 94). That is some extraordinary math. The majority places individual biomechanical engineers on equal footing with the consensus perspective of every major medical society in the world and grants the former a veto over the latter. What’s more, it does so in a case about the admissibility of a medical diagnosis.

As the majority concedes, Dr. Van Ee testified that “he is not a medical doctor, does not have any medical degrees, has not diagnosed or treated patients, has not been taught what a forensic examination for child abuse entails, has never been trained in diagnosing SBS/AHT, has never . . . consulted [on] an examination for abuse . . . and has never examined an infant.” Ante at ____ (slip op. at 56). That is because biomechanical engineers do not diagnose or treat children -- pediatricians, neurologists, radiologists, ophthalmologists, and many other medical professionals, whose professional societies unanimously support SBS/AHT as a valid medical diagnosis, do.

This is not to say that biomechanical engineers can never be relevant to SBS/AHT; rather, that the voices of some biomechanical engineers cannot justify barring evidence that is “wide[ly] support[ed] within the relevant scientific communit[ies]” that actually diagnose children. J.L.G., 234 N.J. at 281.

III.

Finally, I note the sweep of the majority’s decision. Acknowledging a debate and granting a new trial to allow experts to dispute the merits of a particular medical diagnosis is entirely different from prohibiting a trial from taking place at all. The majority has chosen the latter, holding that “expert testimony regarding SBS/AHT without impact . . . cannot be admitted at trial.” Ante at ____ (slip op. at 107).

Under the majority’s rule, unless “new, reliable, scientific evidence is developed . . . that expert testimony regarding SBS/AHT without impact is reliable,” ante at ____ (slip op. at 108), no person in New Jersey can be charged with child abuse for shaking or slamming an infant unless external evidence of impact is present. Similarly, the Division of Child Protection and Permanency (DCPP) will no longer be able to bring Title 9 proceedings in such cases. The Attorney General and DCPP warn that this will be “a significant setback to public safety” and will “hamper DCPP’s ability to fulfill its responsibilities . . .

to ‘immediately take such action as shall be necessary to insure the safety of the child’ upon receiving a report of [child] abuse.” (quoting N.J.S.A. 9:6-8.11).

In my view, a better approach would be to allow a full exchange between experts on the merits of an SBS/AHT diagnosis in any particular case. The majority notes that each defense expert “testified that they believed something other than SBS/AHT could have caused D.J.’s symptoms.” Ante at ____ (slip op. at 104). It observes that Dr. Scheller “stated that he believed D.J. has another condition, hygroma, that mimics SBS/AHT,” and “Dr. Mack testified that the cause of D.J.’s symptoms could be the result of [benign enlargement of the subarachnoid space] or [benign external hydrocephalus].” Ante at ____ (slip op. at 47, 105).¹⁰ They can present those theories at trial. But so should Dr. Medina be able to present her diagnosis and testify as to competing explanations. The outcome of that exchange properly belongs to the jury, not this Court. See, e.g., Lemons, 22 N.W.3d at 62 (disagreements between

¹⁰ The majority underscores Dr. Scheller’s testimony that the number one cause of retinal hemorrhages is “being born normal,” ante at ____ (slip op. at 46), and Dr. Medina’s testimony that “male premature babies are especially prone to subdural hematomas at birth,” ante at ____ (slip op. at 39). But it does not mention that “[s]mall birth-related subdural hematomas . . . resolve in the overwhelming majority of infants within the first 4-6 . . . weeks,” with “all resolved by 3 months,” Consensus Statement at 1057, or that “birth-related retinal hemorrhages” are not multi-layered and the “vast majority . . . resolve by two to four weeks,” Narang, 11 Hous. J. Health L. & Pol’y at 550.

“proponents of the SBS diagnosis as well as experts . . . who disagree with or are skeptical of the SBS diagnosis” “are a matter of weight, not admissibility”).

This case should inspire judicial modesty. It should call to mind Chief Justice Rehnquist’s warning that judges resist the temptation to believe themselves “amateur scientists.” Daubert v. Merrell Dow Pharms. Inc., 509 U.S. 579, 601 (1993) (Rehnquist, C. J., concurring in part and dissenting in part). Yet the majority “step[s] beyond its role as gatekeeper of relevant and reliable information, and instead act[s] as the final arbiter of the correctness of Dr. Van Ee’s” and other experts’ conclusions. Lemons, 22 N.W.3d at 56 (citations omitted). For these reasons, I respectfully dissent.